GROUND·ED DI LLC

DI WEATHER STATION

INSIDE THE WATCH

PUBLIC TECHNICAL EDITION / SELECTIVELY REDACTED

This edition is derived from the complete internal research master. Two protected mathematical subexpressions in the DI-TOR-01 boundary-interaction formula are removed. The original formula and internal receipt remain unchanged.

Published calculations can be independently evaluated only to the extent their inputs and expressions are disclosed. A fully independent computation of the withheld boundary factor is not possible from this public edition.

Scientific status: source-linked event reconstruction and synthetic arithmetic replay; no event-specific tornado score, validated predictive skill, or operational warning authority.

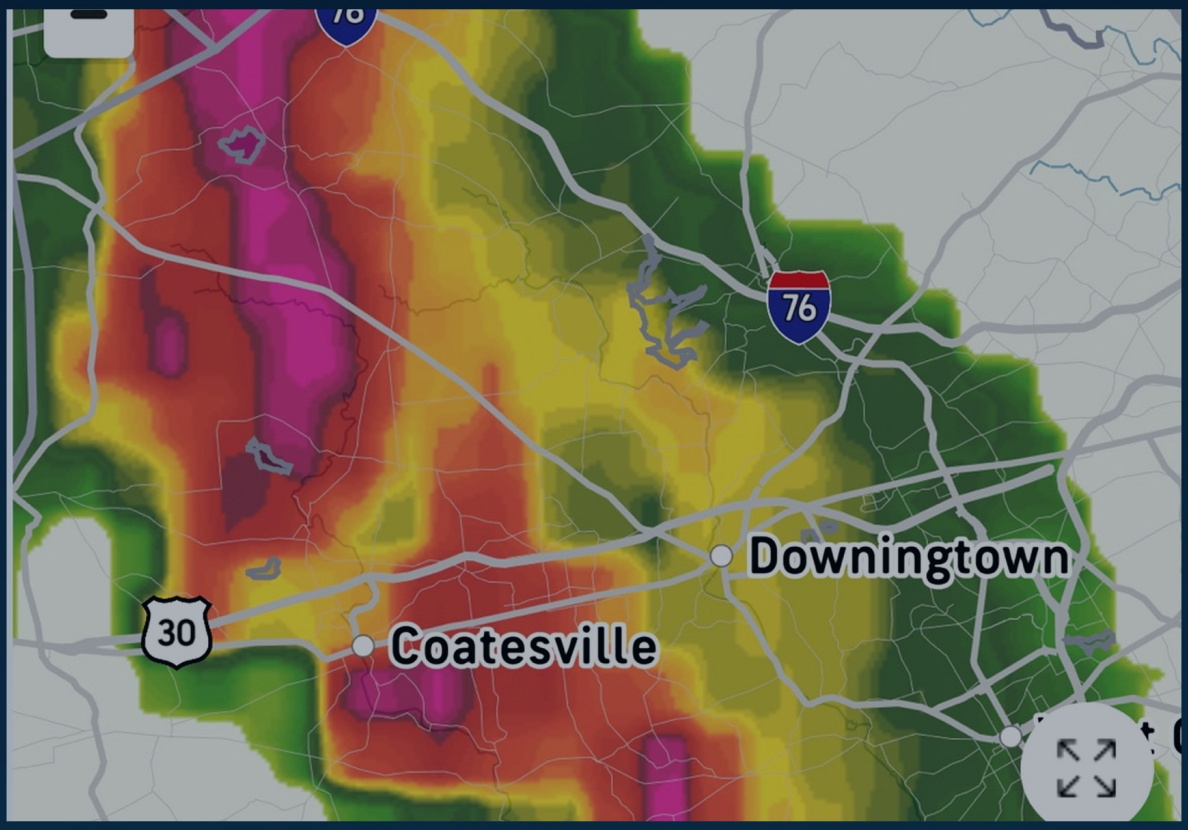
PUBLIC RESEARCH EDITION: Original weather-provider screenshots and field photographs are retained as contextual evidence, with source attribution. Third-party rights are reserved to their owners; publication and onward reuse remain subject to applicable permissions, terms and rights review. Protected boundary-interaction implementation is withheld.

GROUNDED DI / DI WEATHER STATION

INSIDE THE WATCH

WEST CHESTER • PENNSYLVANIA

NOVEMBER 12, 2026



AN EVIDENCE-BOUNDED SEVERE-WEATHER CASE STUDY

GROUNDED DI LLC | COMPLETE UNREDACTED INTERNAL EDITION

THE WEATHER STATION, THE CALCULATION, THE REPLAY.

DI Weather Station / Inside the Watch / West Chester, Pennsylvania

Meteorologist-facing introduction to the public evidence edition | November 12, 2026

What this system actually produced

DI Weather Station is an LLM-assisted weather analysis and replay workflow. For the West Chester case, it reconciled timestamped airport reports, weather-provider data and field context; executed conventional pressure and wind calculations; maintained a versioned research formula; and preserved numerical inputs and executable replay logic. This case demonstrates auditable event reconstruction, not a claim of certified operational forecasting.

EVENT-BASED OUTPUT	REPLAYED RESULT	STATUS
KOQN altimeter trend, 10:15–13:15 EST	−1.8061 hPa/hour	Observed-source arithmetic
KOQN altimeter trend, 12:15–13:15 EST	−2.0318 hPa/hour	Observed-source arithmetic
KOQN wind change, 12:15–13:15 EST	+2 kt; +33.33%; −10°	Observed-source arithmetic
WillyWeather wind change, 13:00–14:00	+1.1 mph; +18.97%; +22.5°	Forecast comparison
DI-TOR-01 Candidate A test	0.15625	Synthetic regression vector
DI-TOR-01 actual event result	No assigned numeric score	Missing matched CAPE and SRH

The contribution being evaluated

The proposition is not that every formula is new. The 0–1 km energy-helicity baseline has published scientific antecedents. The proprietary research contribution under evaluation is the architecture that binds inputs to sources, preserves actual versus synthetic calculations, tests boundary-interaction candidates, and emits machine-readable replay and governance states.

Requested review: assess whether this source-to-calculation replay workflow is useful alongside operational meteorological systems, and identify additional archived inputs needed to evaluate DI-TOR-01 on this event.

Independent numerical replay

Reference event DWS-EVENT-20261112-WC-01. Analytical lock: 18:30 UTC (1:30 PM EST). Airport: KOQN. Source lines are preserved in the accompanying JSON receipt, with source registry and evidence archive.

A. Three-hour altimeter tendency / DI-PRES-001a

10:15 AM — A2954 = 29.54 inHg. 1:15 PM — A2938 = 29.38 inHg.

ΔP = 29.38 − 29.54 = −0.16 inHg; Δt = 3.00 hours.

ΔP / Δt = −0.0533333333 inHg/hour × 33.8638866667 hPa/inHg = −1.8060739556 hPa/hour.

B. One-hour altimeter tendency / DI-PRES-001b

12:15 PM — 29.44 inHg. 1:15 PM — 29.38 inHg.

(29.38 − 29.44) / 1 hour × 33.8638866667 = −2.0318332000 hPa/hour.

C. Airport winds / DI-WIND-001

12:15 PM: wind from 130° at 6 knots. 1:15 PM: wind from 120° at 8 knots.

Δspeed = 8 − 6 = +2 kt; relative change = 100 × 2/6 = 33.3333%; signed circular change = −10° (backing).

D. Forecast comparison / DI-WIND-002

WillyWeather: 1 PM SSW 5.8 mph; 2 PM SW 6.9 mph. Compass centers SSW=202.5° and SW=225°.

Δspeed = +1.1 mph; relative change = 100 × 1.1/5.8 = 18.9655%; direction = +22.5° (veering).

E. Research formula arithmetic / DI-TOR-TEST-001

Synthetic inputs: MLCAPE=500 J/kg, signed $SRH_{0–1}$=100 m^2/s^2, B=0.50.

E01=(500 × max(0,100))/160,000=0.3125. Candidate A=E01 × B=0.3125 × 0.50=0.15625.

This is a test vector, not a measured tornado score for West Chester. The actual event value requires time-matched atmospheric fields.

Machine replay

Run the bundled Python standard-library replay. It reads the preserved JSON inputs, performs decimal arithmetic, checks regression assertions and emits a structured output. Reproduction of these numbers is an arithmetic result; evaluation of forecast skill requires a separate study.

What the system can—and cannot—conclude

EVIDENCE FAMILY	ROLE IN THE CASE	BOUNDARY
KOQN original METAR lines	Source numerical pressure, wind and lightning reports	Airport—not the field camera site
NWS/SPC meteorological products	Official threat context and regional forecast environment	Forecast fields not automatically observed point values
Provider radar/pressure screenshots	Timestamped visual context for the event	Not calibrated Doppler velocity or numerical gradients
User field photographs and videos	Illustration of weather and ground conditions	Not a calibrated rain or wind sensor
DI-TOR-01	Research tornado-potential candidates and sensitivity	Needs event CAPE, signed SRH and boundary inputs
GOEI / evidence control entropy	Proposed source and control governance layer	Numeric control parameters not yet established

Scientific review questions

- Can archived 18:30 UTC RAP/HRRR fields provide time-matched West Chester MLCAPE and 0–1 km signed SRH with documented parcel, storm-motion and spatial-sampling choices?
- How should the storm object and surface boundary be identified to reproduce the bounded boundary-interaction factor across independent cases?
- Does the boundary feature improve discrimination beyond the published modified energy-helicity baseline, using tornado and non-tornado case groups and held-out testing?
- What is the most appropriate radar-derived arrival-time evaluation, including storm-footprint intersection and ground-truth arrival rather than just distance divided by speed?
- Which source-binding and replay artifacts would best support an external weather model audit or governance integration?

Review classification

Completed: observed-source arithmetic, forecast arithmetic, research-formula synthetic regression and source-linked event chronology. Not completed: event-specific DI-TOR calculation; predictive-skill calibration; GOEI event score; operational alert authorization. These states are recorded once here and expanded in the technical appendix of the complete book.

Review files and provenance

This public derivative preserves the original event narrative, source-linked numerical calculations, contextual exhibits and source registry. Protected boundary-interaction implementation details are withheld. The internal research master and original replay receipt remain unchanged.

ARTIFACT	PURPOSE
Inside the Watch original master	Complete original narrative, figures and contextual exhibits
Inside the Watch Replay Receipt.json	Raw METAR lines, formula definitions, results and unresolved fields
Inside the Watch Replay.py	Executable decimal arithmetic and internal test assertions
Source Registry.csv	Identifiable providers, document roles and source URLs
Exhibit Registry.csv	Chronology, photo labels, imagery roles and file hashes
Original Evidence Bundle.zip	Full original event image and video files
DI Weather Station v1.1 update package	Proposed replay, governance and software architecture; release independently gated

Provenance example: the 1:30 PM lock

The 1:15 PM airport record is the nearest listed KOQN pre-lock observation. The 1:35–1:36 PM radar images and later tree, pressure-map and wind-video assets are presented at their contemporaneous times, as a post-lock visual sequence. A screenshot captured at 1:41 PM can still display a weather-map field valid at 1:00 PM; both times are kept without cluttering its caption.

Audience and distribution

Prepared for technical review by atmospheric scientists, operational meteorologists and weather software engineers. Evidence images are retained for transparent source appraisal, not as evidence of calibrated radar velocity or tornado detection. Third-party media rights remain with their respective owners.

Grounded DI LLC | DI Weather Station | StormWise DI² research lineage

PROTOCOL A / DI WEATHER STATION

INSIDE THE WATCH

West Chester, Pennsylvania | November 12, 2026

Public Technical Evidence Edition | Selectively Redacted | November 2026

THE BOOK'S TESTABLE CLAIM

At 1:30 PM EST, the evidence supports a defined tornado-watch context, a measurable airport pressure decline, observed airport winds and lightning, and visually documented heavy precipitation nearby. The supplied inputs do not establish a measured tornado circulation or permit an event-specific DI-TOR-01 score.

Publication identity

Field	Edition information
Publication	DI Weather Station — Inside the Watch
Publisher / brand	Grounded DI LLC / DI Weather Station
Case location	West Chester, Chester County, Pennsylvania
Case date	Thursday, November 12, 2026
Analysis lock	1:30 PM EST / 18:30 UTC
Evidence series	10:15 AM airport observations through afternoon field photographs and video
Runtime posture	FastPath Pro; Tier 20 requested; no certified runtime represented
Research scope	Independent arithmetic replay, source reconciliation, experimental tornado-formula research
Distribution state	Public research derivative; proprietary boundary implementation withheld

Reader's orientation

This edition keeps two things visible at once: a vivid account of an electrically active, rain-heavy afternoon storm, and a calculation record that does not depend on how dramatic the pictures appear. The supplied images are reproduced as contextual exhibits. Meteorological arithmetic is derived separately from recorded observations and published mathematical definitions.

EXECUTIVE INTELLIGENCE

At the analysis lock

On Thursday, November 12, 2026, an official tornado watch included Chester County. The watch was a regional statement of favorable conditions, not a verified tornado at West Chester. A contemporaneous Philadelphia OEM notification appeared at 12:53 PM and stated an expiration of 7:00 PM. A later West Chester alert check independently carried the same county watch through that hour. [S03, S04]

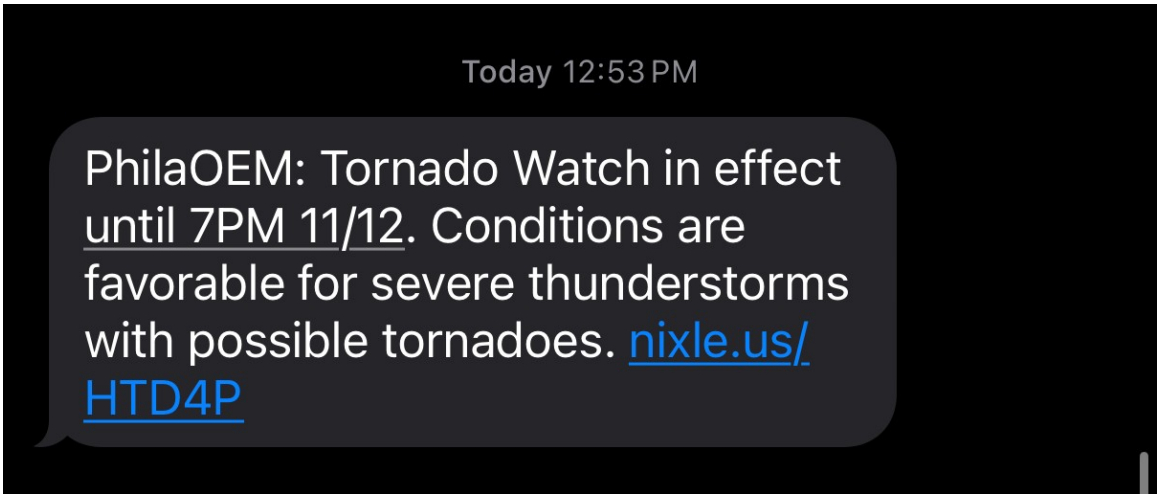


FIG. 01 12:53 PM — Philadelphia OEM watch notification, reproduced in full as a contextual alert exhibit.

Indicator	Research finding at / around lock
Official posture	Tornado watch; no independently issued model warning
Nearest retrieved KOQN report	1:15 PM: A29.38, wind 120° at 8 kt, distant lightning all quadrants
3-hour pressure tendency	−0.16 inHg / 3 h = −1.8061 hPa/h
1-hour pressure tendency	−0.06 inHg / 1 h = −2.0318 hPa/h
Regional NWS forecast	Low instability, modest bulk shear and an opportunity for severe convection
Actual DI-TOR-01	UNRESOLVED: time-matched MLCAPE and 0–1 km SRH unavailable
Synthetic arithmetic test	E01 = 0.3125; Candidate A = 0.15625, explicitly hypothetical
GOEI / H_DI scores	UNRESOLVED, not repurposed as tornado probabilities

EXECUTIVE JUDGMENT

Measured airport pressure decline and photographs document an intense storm episode, not tornadic rotation. DI-TOR-01 remains research-only.

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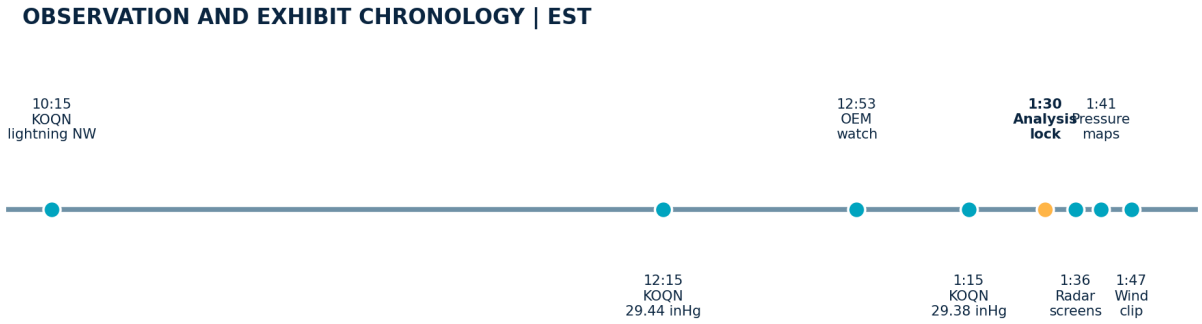


FIG. 02 Chronology of independent airport observations, alert context, and post-lock exhibits. Diagram is a navigation aid, not a radar-derived storm track.

PART 01 / DI WEATHER STATION

THE STORM

The watch, the pressure field, the rain band, and the view from West Chester.

01 / The watch environment

At midday, the relevant official hazard posture was a tornado WATCH. A watch identifies an environment in which tornadic storms are possible within a designated area; it is not evidence that a tornado has been detected at any particular address. The user's 12:53 PM emergency-management message stated a watch through 7:00 PM. The later West Chester weather feed reported Chester County among the covered counties. [S03, S04]

The record does not contain a complete archive of warning polygons active at exactly 1:30 PM. That absence is a source limitation; it must not be converted into proof that no warning could have existed. This case study therefore evaluates conditions and the research formula without issuing an independent tornado warning.

02 / The regional system

The National Weather Service Mount Holly Area Forecast Discussion issued at 2:41 AM EST described a strengthening low-pressure system moving across the Mid-Atlantic and consolidating offshore, with showers and embedded thunderstorms possible during the afternoon. It placed regional forecast convective available potential energy in roughly the 250–500 J/kg range, noted bulk shear around 40 kt, and identified a possible localized convergence zone. [S02]

Those numbers are valuable as an operational forecast of the environment. They are not a 1:30 PM West Chester sounding and, importantly, the bulletin does not supply the specific MLCAPE/SRH pair needed by the candidate tornado equation.

The source-consistent interpretation is a low-instability, shear-bearing severe-convection setup capable of damaging winds and hail. An independently reproduced later SPC outlook described a Slight Risk over a small portion of eastern Pennsylvania and central New Jersey and a possible isolated-supercell component; the present book retains its secondary-source qualification. [S05]

03 / The alert chronology

Time (EST)	Event	Source / analytical class
2:41 AM	Mount Holly regional discussion; modest instability and shear forecast	S02 / OFFICIAL FORECAST

Time (EST)	Event	Source / analytical class
12:50 PM	Reported issuance of Tornado Watch 705	S06 / SECONDARY REPRODUCTION
12:53 PM	Philadelphia OEM watch notification received; stated end 7 PM	EX-001 / CONTEMPORANEOUS ALERT
1:30 PM	DI Weather Station analytical lock	Research analysis setting
1:53–1:57 PM	Later West Chester feed showed Chester County watch through 7 PM	S04 / LATER ALERT CHECK

At the lock, an NWS watch is a valid reason to keep the hazard under review. It is not a numerical tornado-risk threshold for DI-TOR-01. The public safety authority remains the National Weather Service.

04 / The precipitation corridor

The submitted AccuWeather screen captures at 1:35–1:36 PM display a narrow, organized precipitation band stretching across part of eastern Pennsylvania toward the Wilmington area. A closer view places prominent returns west of Downingtown and around Coatesville. The colors and spatial shape provide a useful visual account of heavy precipitation within the broader storm line. [S07]

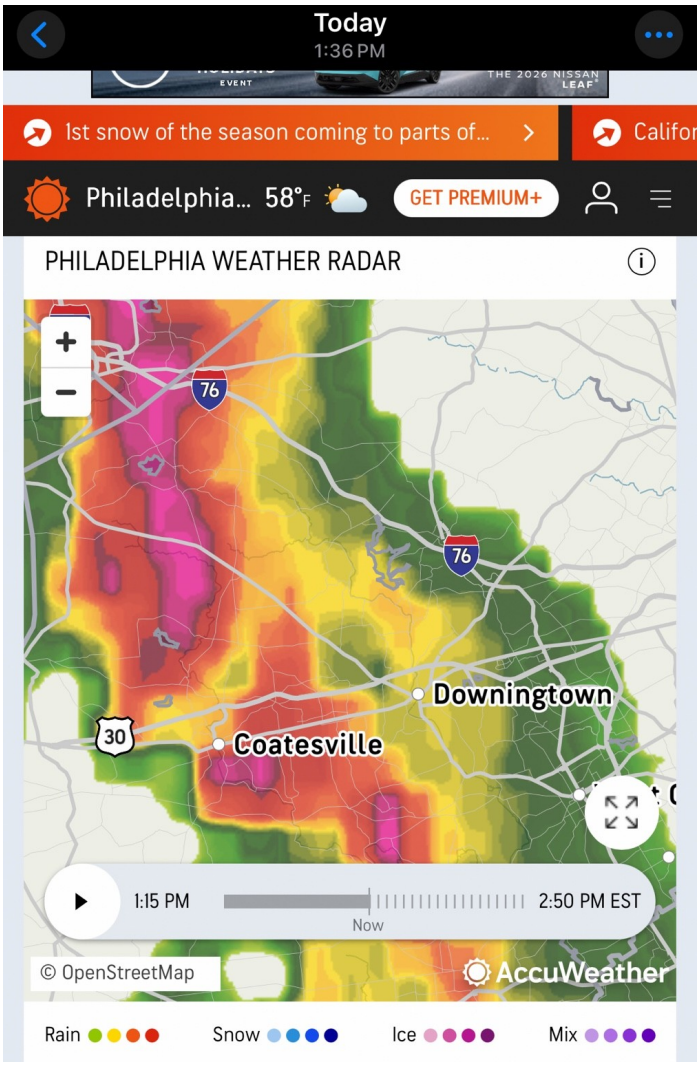


FIG. 03 1:36 PM — Coatesville and Downingtown radar detail. This contextual exhibit documents the displayed precipitation corridor; color alone does not encode tornado rotation.

These captures are not calibrated Doppler-velocity products, and the selected scan time is not independently readable from the playback controls. Accordingly, the book does not calculate a storm velocity, rotation couplet, reflectivity in dBZ, or path vector from this image.

05 / Ground conditions

The submitted field photographs show hard rain striking pavement, water on the parking surface and sidewalks, localized pooling around landscaping, and concentrated drainage along a grassy slope. They make the hydrologic and visual intensity of the event concrete without requiring an invented rain gauge.

A view taken at 1:38 PM shows trees under a broadly overcast sky. At 1:43 PM, a closer image records a tree and nest. A 1:46 PM video screenshot shows conifer foliage. Their value is temporal continuity and context: what the location looked like while the official watch was in force.



FIG. 04 Storm period — parking area rainfall and runoff. Original photograph reproduced without concealment.

06 / The 1:30 PM checkpoint

The analytical lock is 1:30 PM EST, equivalent to 18:30 UTC. The closest recovered KOQN surface record is 1:15 PM, fifteen minutes before the lock. It reports an altimeter setting of 29.38 inHg, easterly-to-east-southeasterly wind from 120° at 8 kt, and distant lightning in all quadrants. [S01]

The lock is a reproducibility boundary. Radar screenshots at 1:35–1:36 PM, pressure maps viewed at 1:41 PM and wind videos around 1:47–1:48 PM belong to the post-lock visual history. They may corroborate the evolving event, but do not masquerade as inputs present at 1:30.

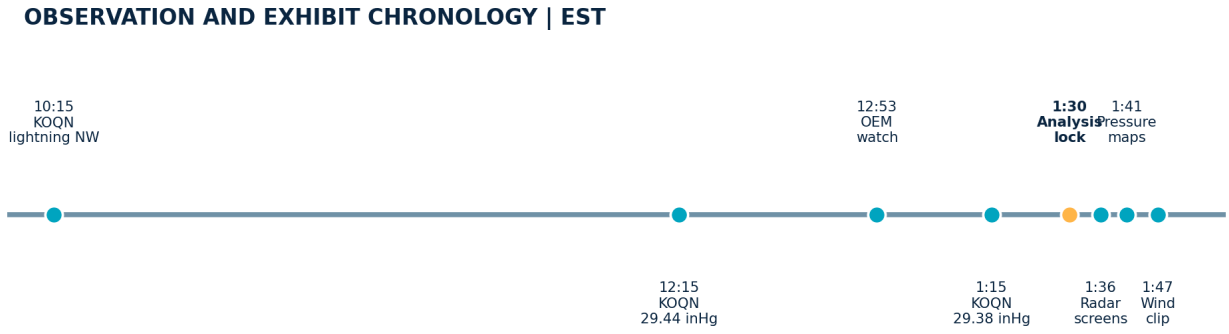


FIG. 05 The observation sequence separates pre-lock measured data from post-lock contextual imagery.

07 / Wind and lightning

The user's account of intense lightning is independently compatible with several KOQN weather reports. The 10:15 AM METAR mentions distant lightning northwest; the 12:15 PM report mentions it north and northeast; and the 1:15 PM report mentions distant lightning in all quadrants. These are airport reports, not claims that the user personally observed lightning at those times. [S01]

Airport wind remained modest in the retrieved reports: 6 kt at 12:15 PM and 8 kt at 1:15 PM, with gusts of 14 kt recorded at 12:35 and 12:55 PM. Those figures do not exclude dangerous convective winds elsewhere within a storm, but they do not themselves prove local tornadic circulation.

08 / The storm unfolds

As the storm moved through the region, the record accumulated three distinct evidence families: the airport's surface measurements, the user's phone weather graphics, and contemporaneous photographs/videos. That combination supports a detailed account of the day while preserving the authority and limitations of each family.

The book's central scientific discipline is therefore simple: calculate from measurement records; use the exhibits to show context, corroboration or contradiction; do not elevate a striking image into a physical measurement it cannot supply.

PART I FINDING

The tornado watch was real, the pressure fall was measurable, and the visual weather was intense. Tornado detection and an event-specific DI-TOR score are separate questions and remain unresolved.

PART 02 / DI WEATHER STATION

THE CALCULATIONS

Source-bound inputs, physical units, independent arithmetic, and replay.

09 / A measurement-first method

The calculation pipeline has five stages: bind a numeric value to a named source; preserve its valid time and location; specify its units and measurement convention; perform arithmetic without borrowing unrelated variables; and verify the result independently. Atmospheric forecasts, surface observations, phone-screen displays, and synthetic research tests occupy distinct classes.

Layer	Question	Allowed role
Observed METAR	What did the airport's instrument report at a stated time?	Actual pressure and wind arithmetic
Official forecast	What environment was expected?	Risk context; cannot be promoted into a point observation
Phone screenshot	What did a provider display when captured?	Context, or a forecast input when the chart value is readable
Field photo/video	What visually occurred around the observer?	Contextual corroboration; quantified only with calibration
Synthetic research vector	Does the candidate equation perform as written?	Arithmetic regression test, not event risk

10 / KOQN atmospheric observations

Brandywine Regional Airport (KOQN), a few miles northeast of West Chester, supplies the quantitative surface backbone. The airport is not the field camera site. Its observations are retained in their original time-stamped METAR encoding in the replay JSON and in the log below. [S01]

Time EST	Altimeter (inHg)	Wind	Reported weather
10:15	29.54	110° / 5 kt	LTG distant NW
12:15	29.44	130° / 6 kt	LTG distant N and NE
12:35	29.42	130° / 4 kt G14	VCTS; LTG distant all quadrants
12:55	29.4	130° / 7 kt G14	light rain; LTG distant SW thru N
13:15	29.38	120° / 8 kt	LTG distant all quadrants

The altimeter group A2938 means 29.38 inHg. LTG DSNT ALQDS means lightning reported distant in all quadrants; VCTS means thunderstorms in the vicinity. None of these abbreviations alone says a tornado occurred.

Selected verbatim raw observation record

15:15Z METAR KOQN 121515Z AUTO 11005KT 10SM BKN025 BKN030 14/10 A2954 RMK AO2 LTG DSNT NW

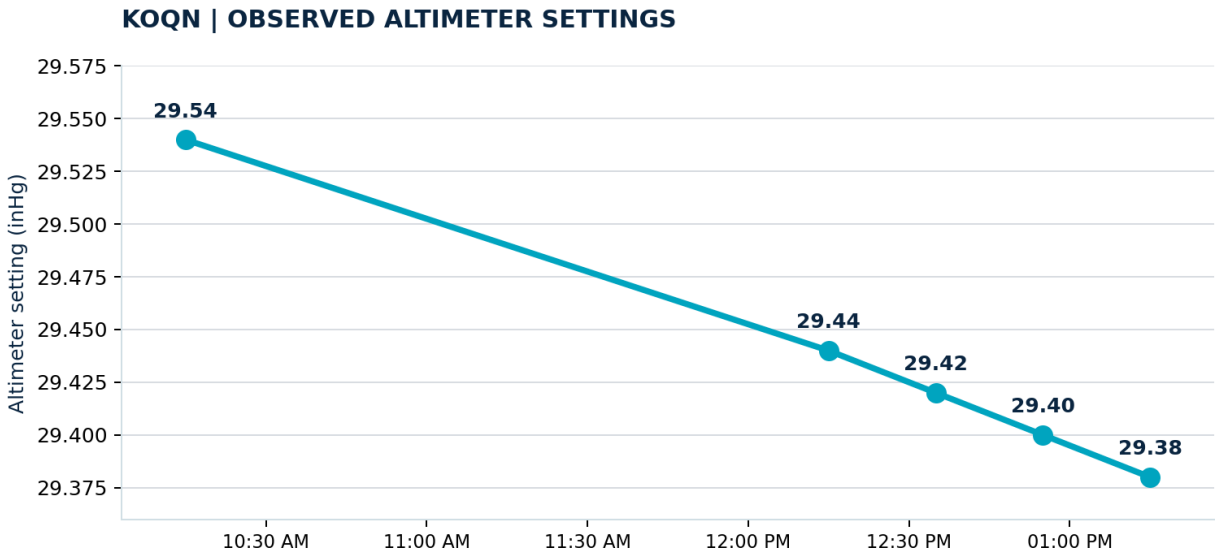
17:15Z METAR KOQN 121715Z AUTO 13006KT 10SM CLR 15/12 A2944 RMK AO2 LTG DSNT N AND NE

17:35Z METAR KOQN 121735Z AUTO 13004G14KT 10SM VCTS CLR 16/12 A2942 RMK AO2 LTG DSNT ALQDS

17:55Z METAR KOQN 121755Z AUTO 13007G14KT 098V168 10SM -RA CLR 16/12 A2940 RMK AO2 LTG DSNT SW THRU N

18:15Z METAR KOQN 121815Z AUTO 12008KT 10SM CLR 15/12 A2938 RMK AO2 LTG DSNT ALQDS

11 / Pressure tendency: a fully worked calculation



Directly reported airport METAR observations; points are not interpolated storm measurements.

FIG. 06 Five timestamped KOQN altimeter settings; the line is a visual connection of the sampled observations.

Pressure tendency is the change in the reported airport altimeter setting divided by elapsed time. It is not a pressure-gradient field and not the central pressure of a tornado.

DI-PRES-001a | Three-hour tendency

$$\Delta P = P(1{:}15\ \text{PM}) - P(10{:}15\ \text{AM}) = 29.38 - 29.54 = -0.16\ \text{inHg}$$

$$\Delta P / \Delta t = -0.16 / 3 = -0.0533333333\ \text{inHg/hour}$$

$$-0.0533333333 \times 33.8638866667 = -1.806074\ \text{hPa/hour}$$

REPLAY RESULT

Three-hour observed altimeter tendency: −1.8061 hPa/h, conventionally rounded to −1.81 hPa/h. The source observations are 10:15 AM and 1:15 PM EST; the elapsed time is exactly three hours.

DI-PRES-001b | One-hour tendency

$$\Delta P = 29.38 - 29.44 = -0.06 \text{ inHg}$$

$$\Delta P / \Delta t = -0.06 / 1 = -0.06000 \text{ inHg/hour}$$

$$-0.06000 \times 33.8638866667 = -2.031833 \text{ hPa/hour}$$

Conversion: one inHg = 33.8638866667 hPa. The unrounded conversion and full numerical inputs are retained in the replay script. A more negative one-hour tendency than the three-hour average is a numerical comparison of two measurement windows, not proof of a tornado circulation.

12 / Wind: observed versus forecast change

DI-WIND-001 | Airport observation change

Between 12:15 and 1:15 PM, KOQN reported a wind speed increase from 6 kt to 8 kt, with wind-from direction 130° to 120°. This is a surface time series, not a vertical wind profile.

$$\Delta V = 8 - 6 = +2 \text{ kt} \qquad \Delta V / V_0 = 2 / 6 = +33.33\%$$

$$\Delta\theta = \text{wrap}(120° - 130°) = -10° \text{ (backing)}$$

The 12:55 PM gust factor is also computable: 14 kt gust divided by a 7 kt recorded mean = 2.00. That one station/time ratio is not a parameterization of tornadic wind speed.

DI-WIND-002 | West Chester forecast comparison

The WillyWeather chart in the submitted 1:47 PM screenshot displays forecast south-southwesterly wind of 5.8 mph at 1 PM and southwesterly wind of 6.9 mph at 2 PM. A compass-center convention translates SSW to 202.5° and SW to 225°. [S09]

$$\Delta V = 6.9 - 5.8 = +1.1 \text{ mph} \quad (+18.9655\%)$$

$$\Delta\theta = 225° - 202.5° = +22.5° \text{ (veering)}$$

The airport's 1:15 PM 120° direction and the 1 PM forecast's 202.5° compass center differ by 82.5° as a circular separation. This comparison crosses locations and product types; it is a discrepancy to investigate, not evidence of spatial convergence or atmospheric helicity.

Calculation	Inputs	Replayed result	Class
Pressure, 3 h	29.54 → 29.38 inHg, 3 h	−1.8061 hPa/h	Observed airport
Pressure, 1 h	29.44 → 29.38 inHg, 1 h	−2.0318 hPa/h	Observed airport

Calculation	Inputs	Replayed result	Class
Airport wind	6 kt,130° → 8 kt,120°	+2 kt; −10°	Observed airport
Gust factor	14 kt / 7 kt	2.00	Observed airport
Forecast wind	5.8 SSW → 6.9 SW	+1.1 mph; +22.5°	Forecast
Lock time offset	1:15 PM → 1:30 PM	15 minutes	Time alignment

13 / Radar and field reconciliation

The 1:35–1:36 PM radar screenshots and the rainfall photographs agree at the qualitative level that an intense precipitation episode affected the surrounding region. The existence of heavy runoff on pavement further supports active rainfall. The evidence does not include calibrated rainfall accumulations, storm-local flash-flood guidance or numeric Doppler velocity. [S07, S10]

The WeatherBug pressure maps captured at 1:41 PM show the 1:00 PM regional pressure field in color. They can illustrate the synoptic-scale pattern, but no pixel-to-pressure calibration and georeferenced gridded values were supplied to compute a local pressure-gradient force. This is not a limitation of the pressure tendency calculation, which uses independent airport METAR values. [S08]

14 / Original StormWise tornado formula

$$\mathbf{TOR_INDEX = CAPE \times SRH \times Boundary\ Interaction}$$

The historical screen gives an approximate 0.3 result with a historical concern reference above 2.0. Its original units, normalization, boundary definition, and calibration sample are not verified against an execution record. The raw product CAPE × SRH has dimensions m^4/s^4 when the usual SI conventions are used. The original historical numerical scale therefore cannot be carried into the new normalized candidate.

15 / DI-TOR-01 v0.2: what can be calculated

The published modified 0–1 km Energy-Helicity Index provides the dimensional baseline used by the new research candidate. The book credits the existing formula and treats the boundary term as the new research hypothesis, not an established predictive improvement. [S11]

$$\mathbf{E_{01} = MLCAPE \times \max(0, SRH_{0\text{–}1}) / 160{,}000}$$

$$\mathbf{Candidate\ A:\ T_A = E_{01} \times B}$$

$$\mathbf{Candidate\ B:\ T_B = E_{01} \times (1 + \beta B)}$$

MLCAPE is measured in J/kg; 0–1 km storm-relative helicity is measured in m^2/s^2, calculated against a specified storm-motion estimate. The 160,000 normalization is in corresponding compound units.

Signed SRH is preserved before any explicitly cyclonic-branch clipping. Boundary factor B and enhancement coefficient β remain research parameters, not operationally established quantities.

DI-TOR-01 | Synthetic arithmetic receipt

$$E_{01} = (500 \times 100) / 160{,}000 = 0.312500$$

$$T_A = 0.312500 \times 0.50 = 0.156250$$

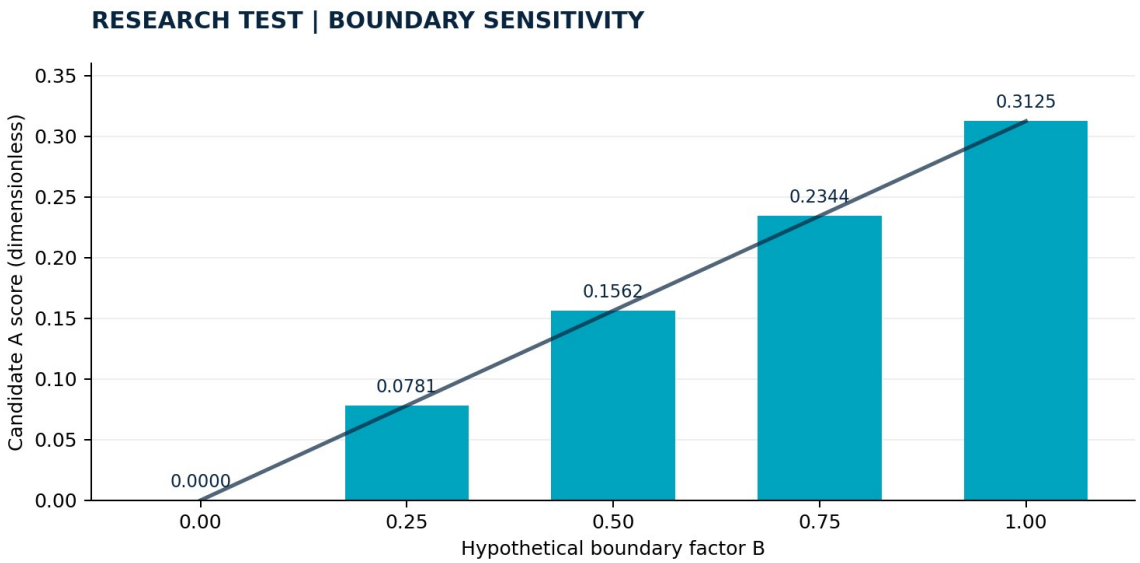


FIG. 07 Synthetic Candidate A response as the hypothetical boundary factor varies from 0 to 1. This graph is not an observed tornado probability.

The actual West Chester event score remains UNRESOLVED because a time-matched local MLCAPE and signed 0–1 km SRH value were not recovered. A synthetic arithmetic pass confirms the equation can be computed from supplied numbers; it does not establish predictive accuracy.

16 / GOEI and governance

$$GOEI = (C \times D \times S) / (R + T)$$

GOEI's historical interpretation uses contextual entropy C, decision complexity D, signal spread S, redundancy R and a threshold constant T. None has a uniquely specified, independently measured numerical value for this event. A hardened runtime variant must define scales and denominator behavior before producing a score. GOEI evaluates the provenance and integrity of an analysis; it does not transform field rainfall into tornado rotation.

$$H_DI = \Sigma_j\, w_j\, [(1 - c_j) + c_j\, h_2((f_j + 0.5)/(n_j + 1))]$$

The evidence-bounded control entropy expression uses control weights, independent coverage fractions, evaluated tests and failure counts. Those configuration and testing inputs were not verified for this event; no H_DI score is asserted.

PART II FINDING

Airport pressure and wind changes are independently reproducible. DI-TOR-01's arithmetic works on explicitly synthetic data, but the actual November 12 score and GOEI/control-entropy values are unresolved. That distinction protects both the mathematics and the meteorological interpretation.

PART 03 / DI WEATHER STATION

THE RESEARCH

From a historical multiplication to a dimensionally normalized, falsifiable weather hypothesis.

17 / Formula lineage, not a novelty claim

StormWise's historical multiplication combined convective available potential energy, storm-relative helicity and an undefined boundary term. It encoded a plausible interaction concept but did not specify units or a reproducible observational interface. In this case study the historical screen remains a lineage artifact rather than a calibrated weather index.

The normalized 0–1 km energy-helicity calculation is already described in an NWS-hosted research review citing Rasmussen (2003). The DI Weather Station contribution under evaluation is the handling of boundary exposure, source provenance, cross-checks and controlled research release—not the invention of the modified EHI baseline. [S11]

18 / Mathematical and physical consistency

CAPE has units J/kg = m^2/s^2. SRH integrated over a specified height layer has units m^2/s^2. Their raw product therefore carries m^4/s^4. The proposed denominator normalizes the product to a dimensionless indicator when those specific input units are used.

The implementation uses the nonnegative portion of signed right-moving-storm 0–1 km SRH to form a research branch focused on cyclonic-supercell environments. Clipping negative SRH values does not imply all storms with negative SRH are safe or incapable of tornadic hazards. The signed source measurement and storm-motion method must be retained.

19 / Boundary interaction: two competing hypotheses

B = [PROPRIETARY BOUNDARY FACTOR WITHHELD]

The bounded boundary-interaction factor B is an experimental, proprietary research construct. Its two protected subexpressions, implementation parameters and calibration rules are intentionally withheld from this public edition. No event-specific B has been established.

Independent evaluation requires separately acquired, time-aligned meteorological observations and a documented evaluation protocol. The public record does not permit independent computation of the withheld boundary factor.

Hypothesis	Equation	Zero-B behavior	Key failure risk
A / attenuation	$T_A = E_{01}\ B$	$T_A = 0$	Suppresses genuine hazard when a boundary is absent, missed or irrelevant
B / enhancement	$T_B = E_{01}(1 + \beta B)$	$T_B = E_{01}$	Can double-count boundary-generated shear/instability; β uncalibrated
Baseline	E_{01} alone	Unaffected	May miss mechanistic differences among supercell, QLCS and other modes

20 / Sensitivity and transparent synthetic testing

For positive inputs and fixed B, Candidate A responds linearly to MLCAPE, 0–1 km SRH and B. Doubling any one doubles its score. This algebraic property can be verified without pretending the relationship is a calibrated tornado probability. Candidate B also responds linearly to the CAPE–SRH baseline but has additive baseline preservation in B.

MLCAPE	$SRH_{0–1}$	Hypothetical B	E_{01}	T_A
500	100	0.00	0.312500	0.000000
500	100	0.25	0.312500	0.078125
500	100	0.50	0.312500	0.156250
500	100	0.75	0.312500	0.234375
500	100	1.00	0.312500	0.312500
1,000	150	0.50	0.937500	0.468750
1,500	200	0.75	1.875000	1.406250

All numbers in this table are synthetic. No tornado-risk threshold is placed on these values, and the historical 0.3 and >2.0 threshold are not imported. The only validated property here is numerical arithmetic for a defined equation.

21 / Missing event-time atmospheric inputs

Required input	Current state	Why it changes computation
West Chester MLCAPE at 18:30Z	UNRESOLVED	Regional forecast CAPE is not a local point/time parcel calculation
Signed 0–1 km SRH	UNRESOLVED	Needs vertical wind profile and a storm-motion vector
0–3 km SRH	UNRESOLVED	Required for conventional-layer EHI comparator

Required input	Current state	Why it changes computation
Effective bulk shear	UNRESOLVED	NWS 40 kt regional forecast not a point/version-locked effective-layer field
MLLCL and CIN	UNRESOLVED	Needed for version-specified STP evaluation
Surface convergence B	UNRESOLVED	Protected boundary inputs withheld; no measured B
Doppler velocity	UNRESOLVED	User radar images are precipitation displays, not verified velocity fields
Historical warning polygons	UNRESOLVED	Watch verified; full time-locked warning history not independently recovered

The data gaps are specific. A time-stamped RAP/HRRR analysis sounding or SPC mesoanalysis grid cell can provide conditional environmental parameters; archived radar velocity and storm-object tracks can evaluate actual rotation and mode. Any spatial interpolation, forecast surrogate or model assumption must be explicitly documented and not labeled a direct observed sounding.

22 / Competing meteorological mechanisms

An intense precipitation band is consistent with several storm modes. Linear or quasi-linear convection can produce strong rain and wind and can sometimes produce tornadoes. Discrete or embedded supercells pose a different rotation-diagnosis problem. A non-tornadic convective line can also be visually dramatic. The supplied radar screenshots do not resolve these alternatives because their Doppler-velocity signatures and storm-cell tracking are unavailable.

Lightning intensity and rapid surface runoff should be examined through parallel modules. A lightning-jump feature needs timestamped flash rates attributable to a tracked storm; a flash-flood guidance ratio needs measured precipitation accumulation and locally matched guidance. Neither can be inserted as a tornado multiplier solely because it co-occurred with the watch.

23 / A credible validation program

Predictive validation needs independent, documented samples of tornado-producing and non-tornado storms, with event-grouped holdouts to reduce leakage. It should compare the baseline E_{01}, Candidate A, Candidate B, standard EHI layer variants and a precisely versioned significant tornado parameter (STP). Stratify by storm mode and geographic regime.

For each candidate, test discrimination and calibration, probability reliability if a probability model is later built, false-alarm rates, missed events, time-to-detection and robustness to boundary-location errors. Do not choose a decision threshold using the same events used to optimize the model.

Test family	Question	Minimum acceptance evidence
Arithmetic	Does the code reproduce formula examples?	Stored regression vectors pass
Input binding	Does each field mean what the equation expects?	Unit, parcel, layer, time and source validation
Incremental skill	Does B improve over E_{01} without leakage?	Out-of-sample comparison over independent cases
Calibration	Does a score map to stable empirical frequencies?	Held-out reliability and uncertainty analysis
Operational safety	Does the system handle missing/conflicting inputs?	Explicit abstention and authority-bound release gates

24 / What the event establishes—and does not

The West Chester case demonstrates a replayable airport pressure analysis, a documented watch context, and a sequence of storm images that align qualitatively with a rain-heavy afternoon. It also demonstrates a disciplined limit: genuine weather intensity in the field is not enough to infer the vertical shear and storm-relative helicity required for DI-TOR-01.

It does not show a verified tornado at the user location, an actual event DI-TOR score, a calibrated tornado probability, independent superior prediction, or certified operational override execution.

25 / Research conclusion

SCIENTIFIC DISPOSITION

Historical formula: reconstructed. Modified EHI baseline: existing scientific convention. DI-TOR-01 v0.2: explicit and arithmetically testable research candidate. Event-specific numerical tornado score: unresolved. GOEI/control entropy: governance concepts without scored inputs. WeatherReady and AlertRelease: not validated / blocked.

The durable product is the analytical record: sources, formulas, time lock, observed measurements, mathematical substitutions, numerical outputs, known contradictions and specified missing inputs. Those are the elements another analyst can replay and contest.

PART 04 / DI WEATHER STATION

THE COMPLETE RECORD

Chronology, unaltered source assets, calculator receipts, and an independent replay specification.

26 / Chronology and evidence lock

Time EST	Record	Interpretation / linkage
10:15 AM	KOQN A29.54; 110°/5 kt; distant lightning NW	Pre-lock airport datum; S01
12:15 PM	KOQN A29.44; 130°/6 kt; distant lightning N and NE	Pressure and wind baseline
12:35 PM	KOQN A29.42; 130°/4 kt gust 14; VCTS	Airport convective vicinity observation
12:50 PM	Reported Tornado Watch 705 issuance	Secondary SPC reproduction; S06
12:53 PM	OEM text screenshot	Contemporaneous alert exhibit EX-001
12:55 PM	KOQN A29.40; 130°/7 kt gust 14; light rain	Airport rain/gust record
1:00 PM	Pressure-map valid time; WillyWeather wind forecast point	Underlying weather-product times
1:15 PM	KOQN A29.38; 120°/8 kt; distant lightning ALQDS	Nearest retrieved pre-lock METAR
1:30 PM	DI Weather Station analysis lock	18:30 UTC
1:35–1:36 PM	Radar screenshots	Regional/local precipitation gallery
1:38 PM	Tree canopy photo	Post-lock visual evidence
1:41 PM	Pressure map gallery captures	1:00 PM data field displayed later
1:43 PM	Tree-and-nest photo	Post-lock visual evidence
1:46 PM	Conifer video frame	Post-lock visual evidence
1:47–1:48 PM	Wind forecast screenshot and field videos	Wind context, not calibrated on-site sensor
1:53–1:57 PM	Later alert feed checks; storm reported	Post-lock status, not pre-lock historical polygon

27 / Analytical source register

The source IDs identify what the independent calculator is allowed to use. A user exhibit can be vivid and timely without possessing the numerical field required by a formula. The full registry, including URLs, is supplied as CSV.

ID	Source family	Role and evidentiary class
S01	KOQN METAR sequence, Nov 12 2026	OBSERVED / SECONDARY ARCHIVE OF PRIMARY METAR
S02	NWS Mount Holly AFD, 02:41 AM EST Nov 12	OFFICIAL FORECAST
S03	Philadelphia OEM watch text	ALERT / USER-SUPPLIED SCREENSHOT
S04	NWS Chester County watch status	OFFICIAL ALERT / PREVIOUS RUN RECORD
S05	SPC November 12 Slight Risk update	SECONDARY / NOT VERIFIED PRIMARY
S06	Tornado Watch 705 text	SECONDARY / NWS ALERT CORROBORATES SCOPE
S07	AccuWeather precipitation screenshots	CONTEXTUAL EXHIBIT
S08	WeatherBug pressure-color maps	CONTEXTUAL EXHIBIT
S09	WillyWeather NOAA wind forecast screenshot	FORECAST SCREENSHOT
S10	Field photographs and videos	CONTEXTUAL EXHIBIT
S11	NWS/NOAA near-storm environment parameter review	SCIENTIFIC REFERENCE
S12	StormWise Final Event Scan	INTERNAL ANALYTICAL RECORD
S13	StormWise machine-readable calculation receipt	INTERNAL REPLAY RECORD
S14	DI Weather Station Hurricane Polo rolling handoff v2	VISUAL/STRUCTURAL REFERENCE
S15	DI Weather Station Regional Showcase	VISUAL/STRUCTURAL REFERENCE
S16	Grounded DI Weather formula packet and runtime packages	HISTORICAL FORMULA/CONTROL REFERENCE

28 / Complete calculation receipts

DI-PRES-001a | KOQN 3-hour altimeter tendency

Receipt field	Replay value
Source / inputs	KOQN A2954 at 10:15 EST; A2938 at 13:15 EST
Equation	(P2 − P1) / 3 h
Substitution	(29.38 − 29.54) / 3 = −0.05333333 inHg/h
Result	−1.8060739556 hPa/h (−1.81 rounded)
Disposition	Observed airport arithmetic / PASS

DI-PRES-001b | KOQN 1-hour altimeter tendency

Receipt field	Replay value
Source / inputs	KOQN A2944 at 12:15 EST; A2938 at 13:15 EST
Equation	(P2 − P1) / 1 h
Substitution	(29.38 − 29.44) / 1 = −0.06 inHg/h
Result	−2.0318332000 hPa/h (−2.03 rounded)
Disposition	Observed airport arithmetic / PASS

DI-WIND-001a | KOQN sustained wind time change

Receipt field	Replay value
Source / inputs	KOQN: 130°/6 kt at 12:15; 120°/8 kt at 13:15
Equation	ΔV = V2−V1; % = 100ΔV/V1; circular Δθ
Substitution	8−6 = 2 kt; 100×2/6 = 33.333%; 120−130 = −10°
Result	+2 kt, +33.33%, −10° backing
Disposition	Observed airport arithmetic / PASS

DI-WIND-001b | KOQN 12:55 PM gust factor

Receipt field	Replay value
Source / inputs	KOQN 12:55: 7 kt sustained, gust 14 kt
Equation	Gust factor = Vgust / Vmean
Substitution	14 / 7 = 2.0
Result	2.00 (unitless)
Disposition	Observed airport arithmetic / PASS

DI-WIND-002 | WillyWeather wind forecast change

Receipt field	Replay value
Source / inputs	1 PM 5.8 mph SSW (202.5°); 2 PM 6.9 mph SW (225°)
Equation	ΔV, 100ΔV/V1, circular Δθ
Substitution	6.9−5.8=1.1; 100×1.1/5.8=18.9655%; 225−202.5=22.5°

Receipt field	Replay value
Result	+1.1 mph, +18.97%, +22.5° veering
Disposition	Forecast arithmetic / PASS

DI-TIME-001 | Age of nearest KOQN observation

Receipt field	Replay value
Source / inputs	Report 1:15 PM; analysis lock 1:30 PM
Equation	t_lock − t_obs
Substitution	13:30 − 13:15 = 15 minutes
Result	15 minutes
Disposition	Time alignment / PASS

DI-TOR-TEST-001 | DI-TOR-01 synthetic regression

Receipt field	Replay value
Source / inputs	MLCAPE=500 J/kg; signed SRH=100 m^2/s^2; B=.50
Equation	E01=MLCAPE*max(0,SRH)/160000; T_A=E01*B
Substitution	500*100/160000 = 0.3125; 0.3125*0.50 = 0.15625
Result	E01=0.3125; T_A=0.15625
Disposition	Synthetic arithmetic / PASS; NOT event data

Exact unrounded decimal output, input JSON fields and executable verification checks are preserved in the accompanying Python replay script.

29 / Formula registry and control state

ID	Formula or construct	Status
Legacy TOR	CAPE × SRH × Boundary Interaction	Historical and numerically uncalibrated
E01	MLCAPE × positive 0–1 km SRH / 160000	Published research baseline, requires event inputs
DI-TOR-A	E01 × B	Research hypothesis; synthetic arithmetic PASS
DI-TOR-B	E01 × (1 + βB)	Research hypothesis; β uncalibrated
DI-RAIN-01	max(P_tau / FFG_tau) for matched intervals	Candidate; no measured local

ID	Formula or construct	Status
		accumulation/FFG
GOEI	C×D×S/(R+T)	Governance research; missing bound inputs
H_DI	Evidence-bounded weighted control entropy	No complete coverage/test-count inputs
Gatekeeper	Validation and AlertRelease state	NOT VALIDATED / BLOCKED

30 / Discrepancies, limitations and state integrity

Issue	Treatment
Regional forecast CAPE vs point MLCAPE	Do not substitute the 250–500 or secondary 500–900 range into the event formula.
WillyWeather forecast vs KOQN wind	Compare but do not equate different locations, products or valid times.
1:41 PM screen vs 1:00 PM pressure map	Use 1:41 as screenshot chronology; 1:00 as underlying data-valid time.
Radar reflectivity vs Doppler velocity	Radar colors support precipitation-band context; rotation remains unverified.
Original 0.3 / threshold 2.0	Retain historically; no transfer to normalized index.
GOEI numerator and denominator	No normalized runtime inputs or denominator constants; abstain numerically.
Warning state at 1:30 PM	Official watch established; complete historical tornado-warning polygon inventory missing.
Missing camera/phone EXIF	Gallery times used in ordinary captions; raw file metadata retained in registry.

REPLAY CONTRACT

Versioned measured inputs + locked formulas + specified units and conversion constants → the same numeric output. A change in source records, formulas, scope or assumptions must produce a new receipt and an explicit explanation, not a silent rewrite.

COMPLETE CONTEXTUAL EXHIBIT ATLAS

31–32 / Photographs, radar and pressure maps

These pages reproduce the original uploaded image files without redactions or selective concealment. Each caption adopts the contemporaneous screenshot/gallery time. Weather-product data-valid times, source classes, dimensions, and SHA-256 hashes are consolidated in the adjacent CSV and file-level audit registry.

The visual exhibits are not the mathematical input for KOQN pressure tendency, and their imagery alone cannot resolve storm-relative helicity or tornado rotation.

ATLAS CONVENTION

Time printed in each caption refers to the contemporaneous image/gallery context when available. For displayed weather graphics, the source's forecast/observation valid time is separately retained in the exhibit registry. Exact, source-identified measurements—not aesthetic intensity—govern formula arithmetic.

CONTEXTUAL EXHIBIT / EX-001 / ORIGINAL ASSET

12:53 PM · Philadelphia OEM tornado-watch notification

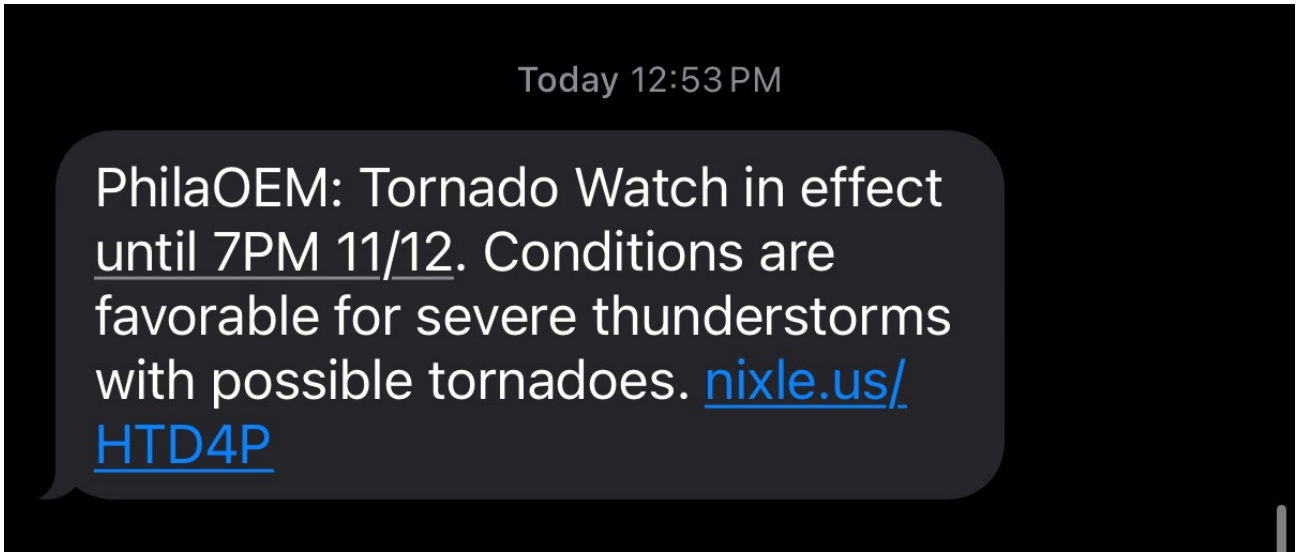


FIG. 08 12:53 PM — Philadelphia OEM tornado-watch notification. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User screenshot / alert	IMG_6542.jpeg	SHA-256 9904a351efd0f20d5ba6…

CONTEXTUAL EXHIBIT / EX-002 / ORIGINAL ASSET

1:35 PM · Regional radar overview

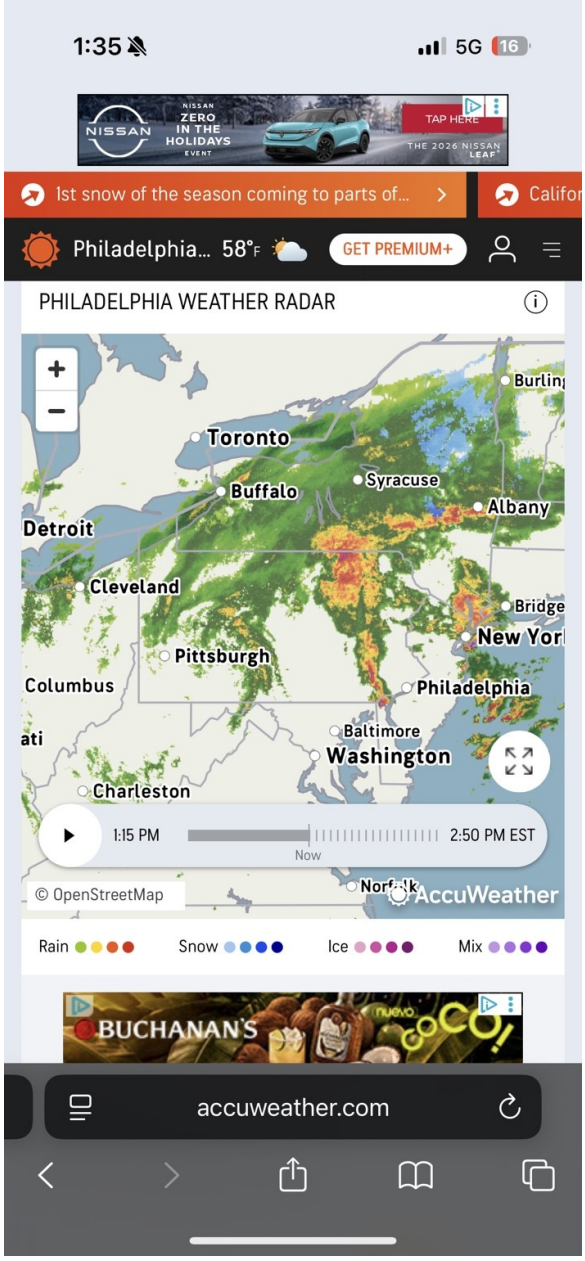


FIG. 09 1:35 PM — Regional radar overview. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
AccuWeather / screenshot	IMG_6547.jpeg	SHA-256 b4e9648511f52d83f49c…

CONTEXTUAL EXHIBIT / EX-003 / ORIGINAL ASSET

1:36 PM · Regional radar corridor

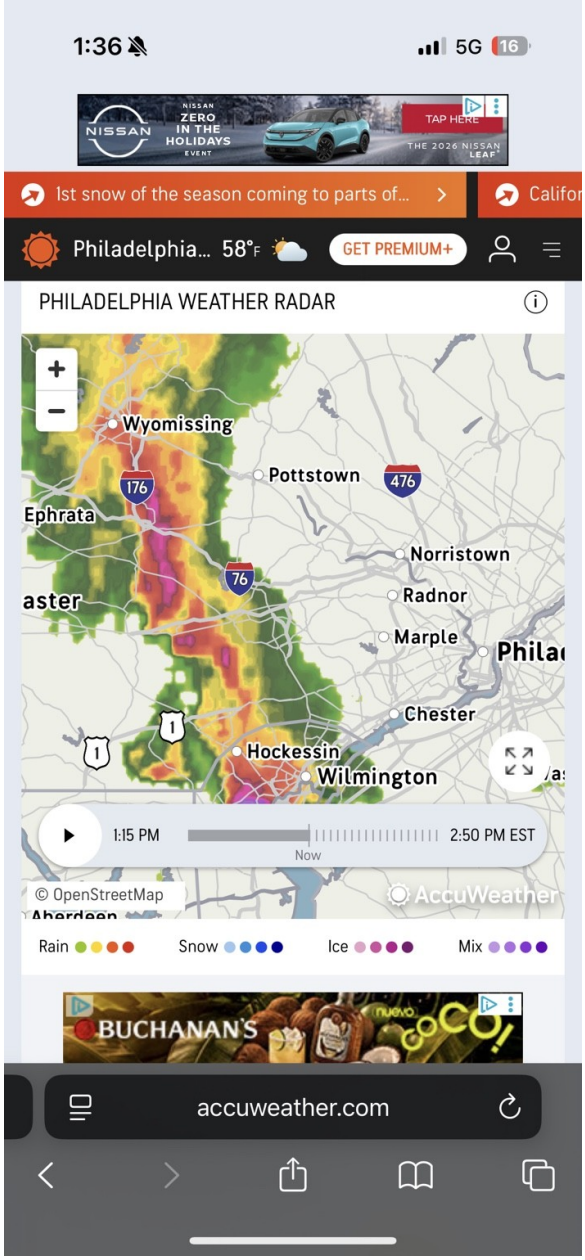


FIG. 10 1:36 PM — Regional radar corridor. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
AccuWeather / screenshot	IMG_6548.jpeg	SHA-256 443d19d25e73e66d558c…

CONTEXTUAL EXHIBIT / EX-004 / ORIGINAL ASSET

1:36 PM · Coatesville and Downingtown radar

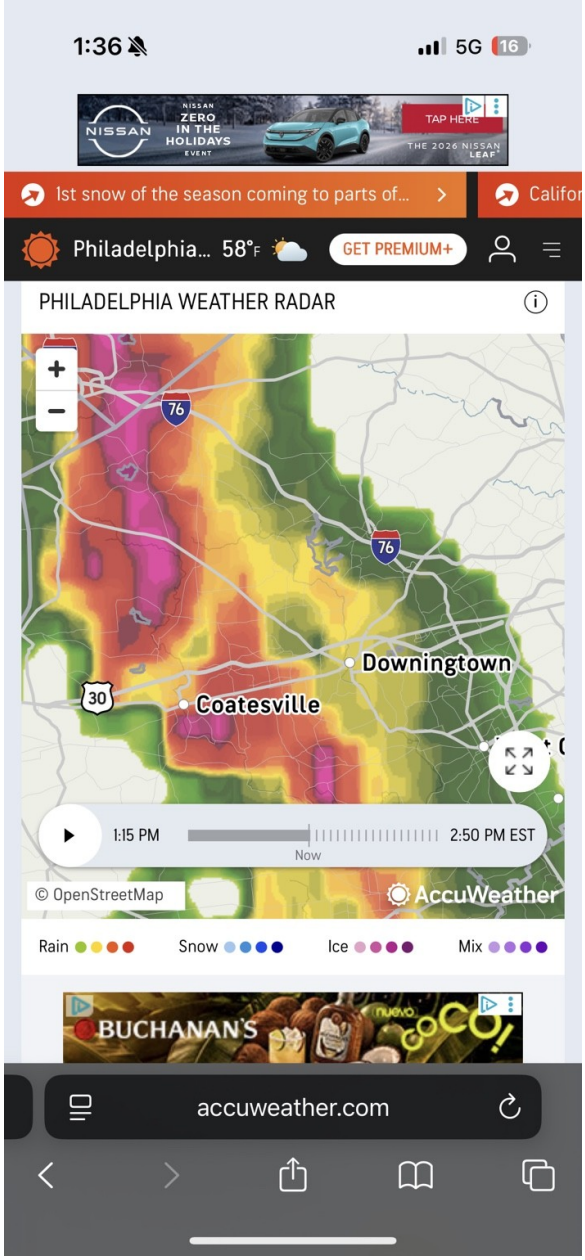


FIG. 11 1:36 PM — Coatesville and Downingtown radar. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
AccuWeather / screenshot	IMG_6549.jpeg	SHA-256 3107b90ad9d647bcad65…

CONTEXTUAL EXHIBIT / EX-005 / ORIGINAL ASSET

1:39 PM · Washington Square West conditions



FIG. 12 1:39 PM — Washington Square West conditions. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
Weather Underground / screenshot	IMG_6552.png	SHA-256 0616abc1498c232ea798…

CONTEXTUAL EXHIBIT / EX-006 / ORIGINAL ASSET

1:41 PM · Northeast regional pressure map

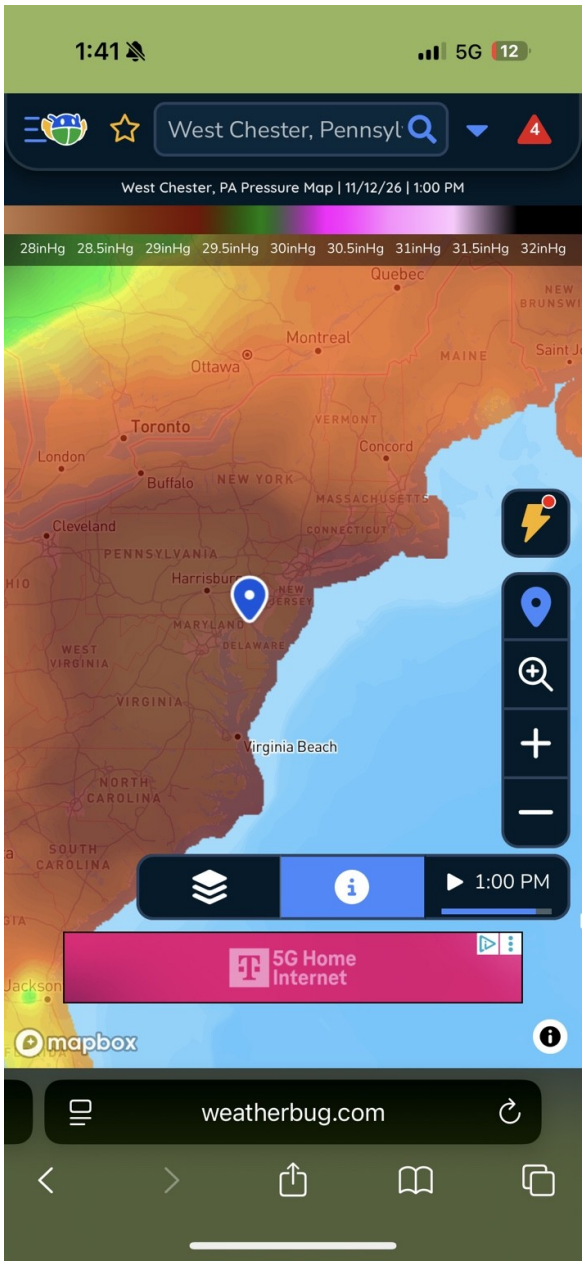


FIG. 13 1:41 PM — Northeast regional pressure map. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6553.jpeg	SHA-256 20c362b520f25a649182…

CONTEXTUAL EXHIBIT / EX-007 / ORIGINAL ASSET

1:41 PM · North America pressure map

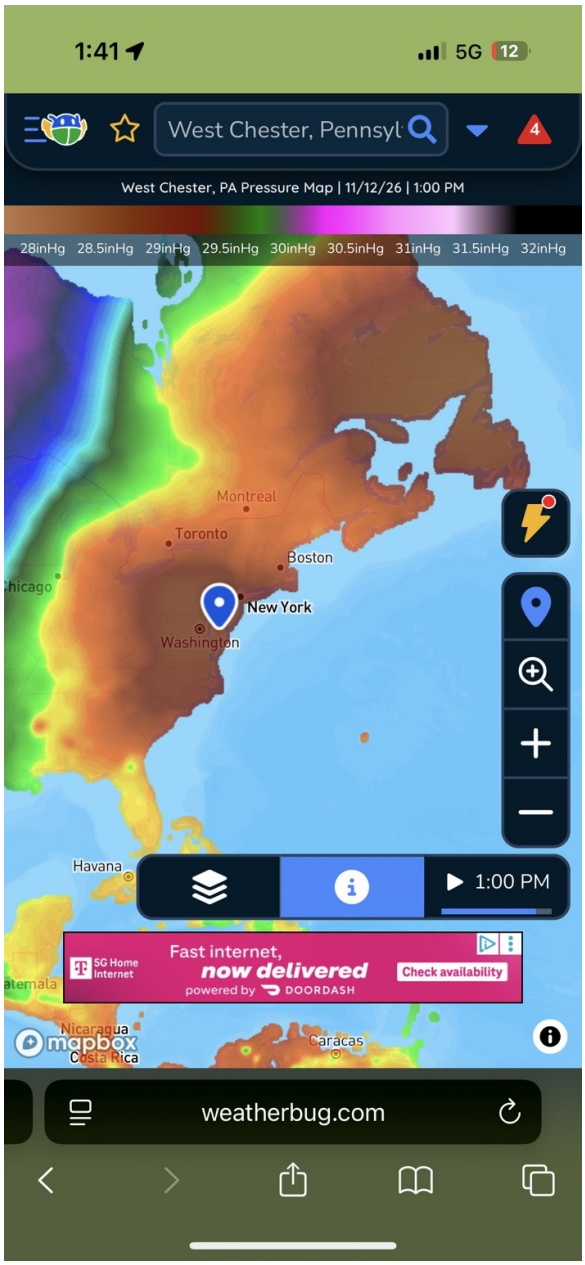


FIG. 14 1:41 PM — North America pressure map. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6554.jpeg	SHA-256 e17f935fea708614b24e…

CONTEXTUAL EXHIBIT / EX-008 / ORIGINAL ASSET

1:41 PM · West Chester pressure map

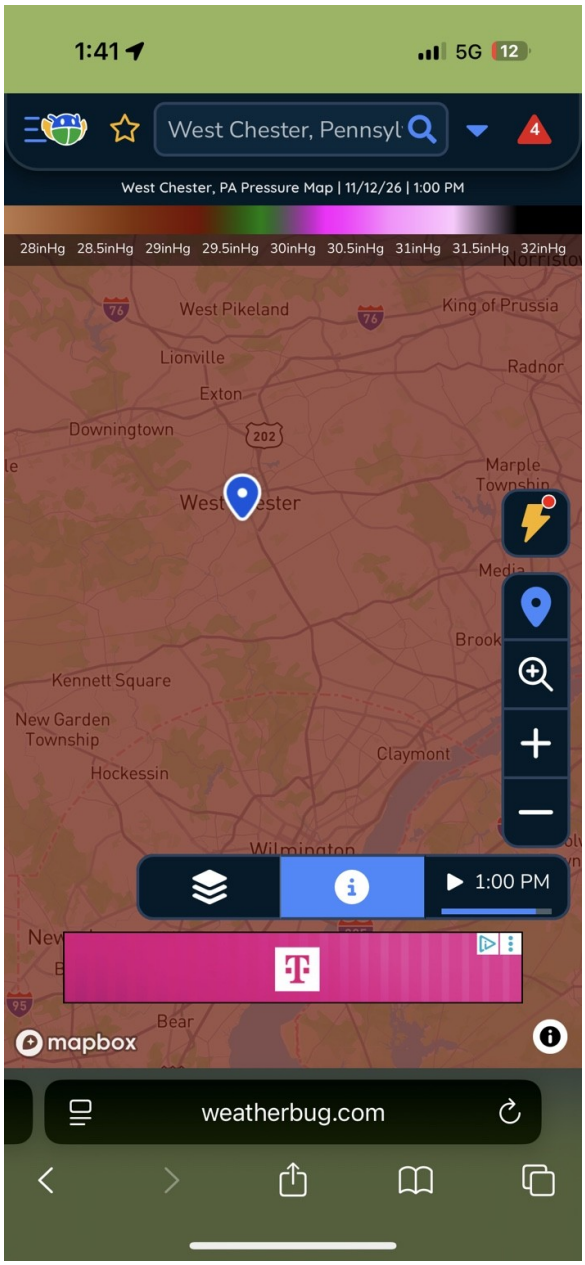


FIG. 15 1:41 PM — West Chester pressure map. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6555.jpeg	SHA-256 24c0d5e92d12a9106c4c…

CONTEXTUAL EXHIBIT / EX-009 / ORIGINAL ASSET

1:41 PM · Regional pressure detail

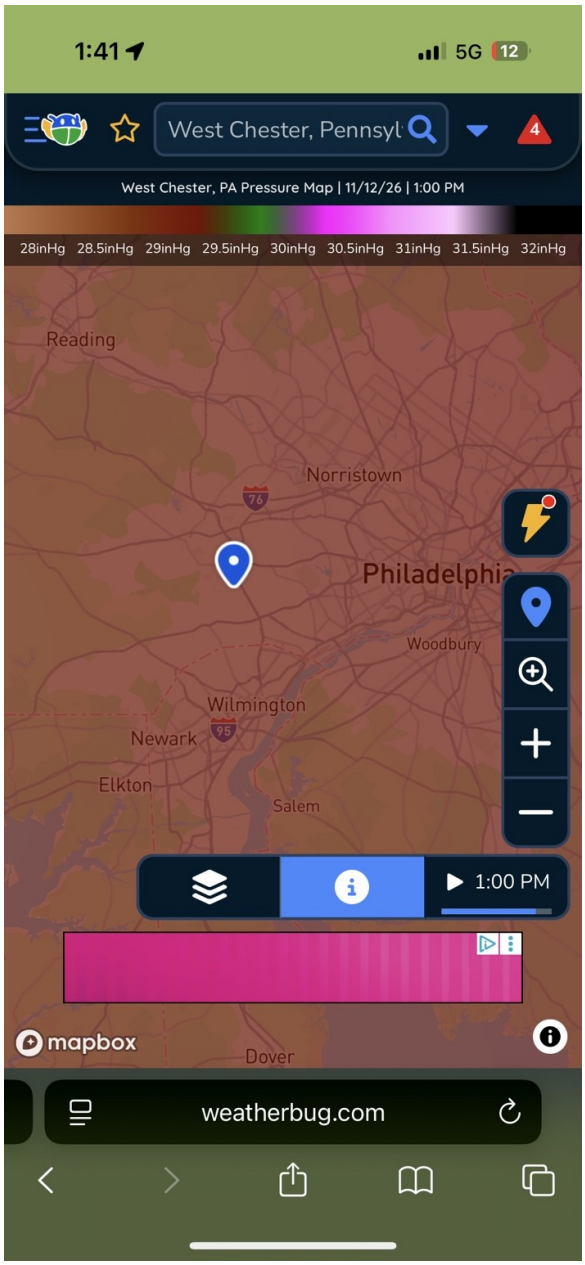


FIG. 16 1:41 PM — Regional pressure detail. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6556.jpeg	SHA-256 71bc468e6d0e0913784b…

CONTEXTUAL EXHIBIT / EX-010 / ORIGINAL ASSET

1:41 PM · Greater Philadelphia pressure

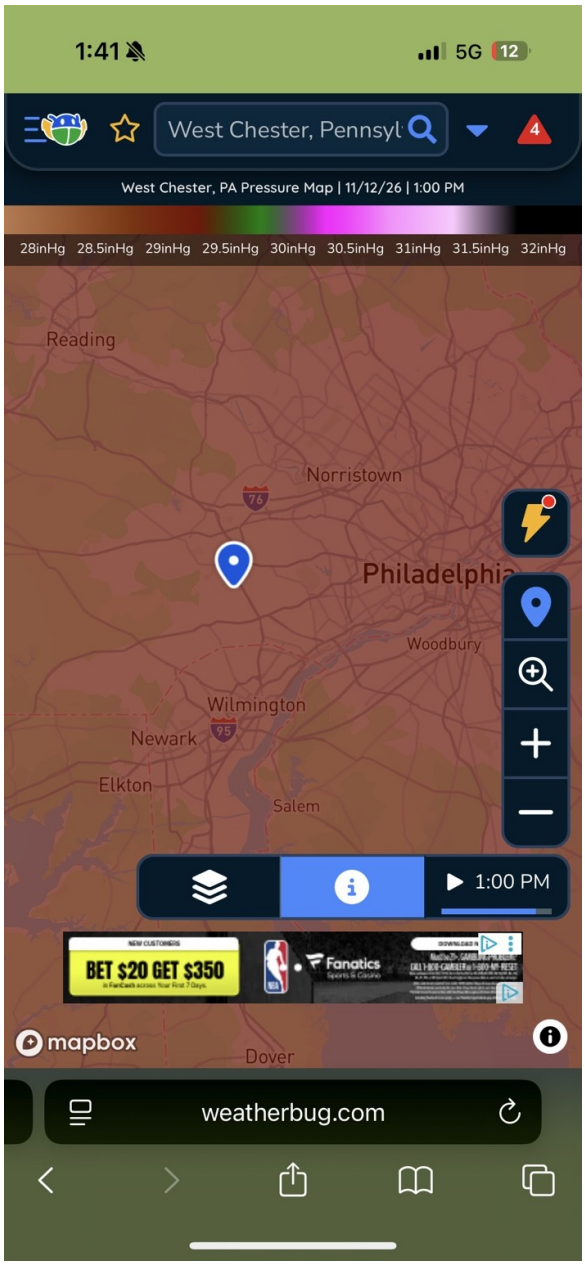


FIG. 17 1:41 PM — Greater Philadelphia pressure. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6557.jpeg	SHA-256 1e7763e72bba7cf4cb7d…

CONTEXTUAL EXHIBIT / EX-011 / ORIGINAL ASSET

1:41 PM · Broader pressure gradient

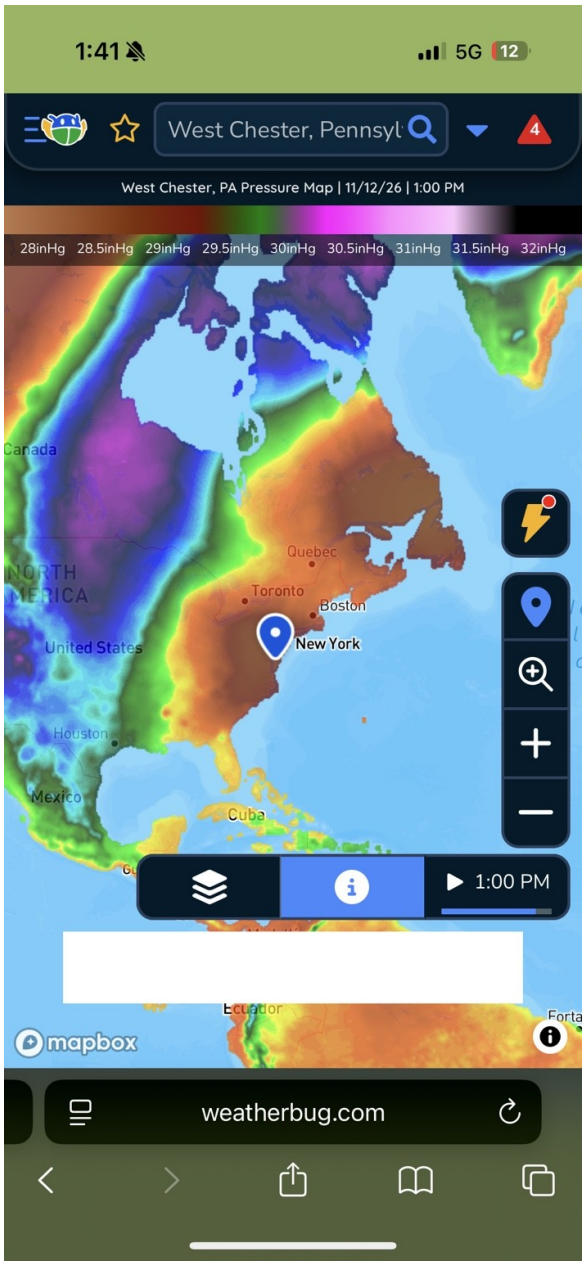


FIG. 18 1:41 PM — Broader pressure gradient. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6558.jpeg	SHA-256 52b899bfdb5f82e19400…

CONTEXTUAL EXHIBIT / EX-012 / ORIGINAL ASSET

1:47 PM · Wind forecast 1 PM

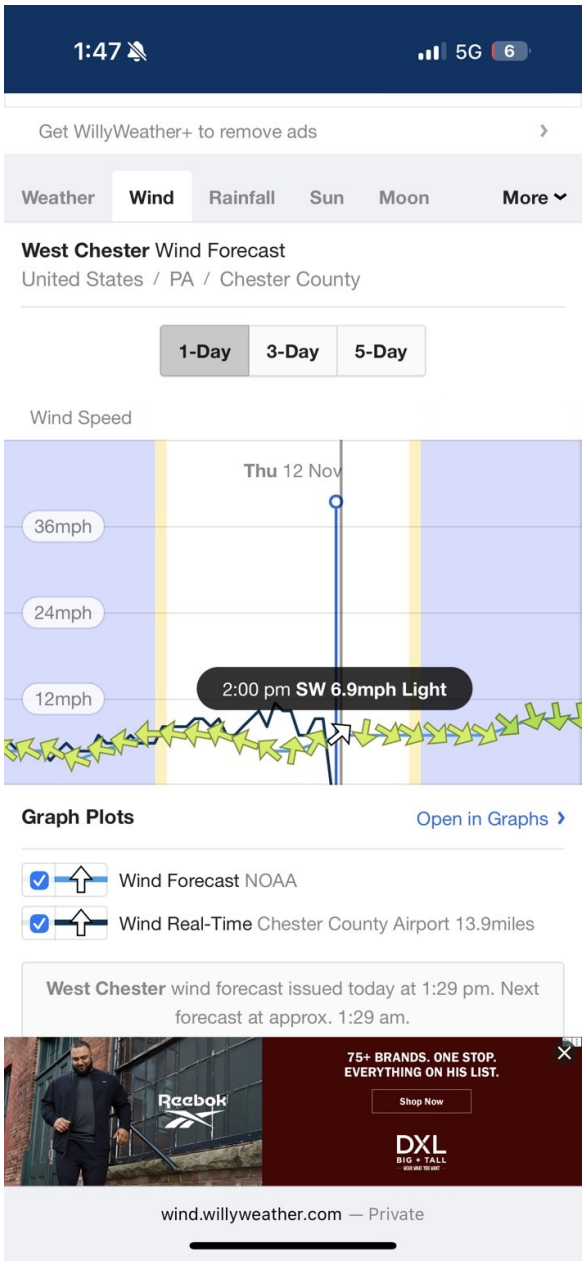


FIG. 19 1:47 PM — Wind forecast 1 PM. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WillyWeather / screenshot	IMG_6563.jpeg	SHA-256 0340531e0c9d8cd19129…

CONTEXTUAL EXHIBIT / EX-013 / ORIGINAL ASSET

1:47 PM · Wind forecast 2 PM

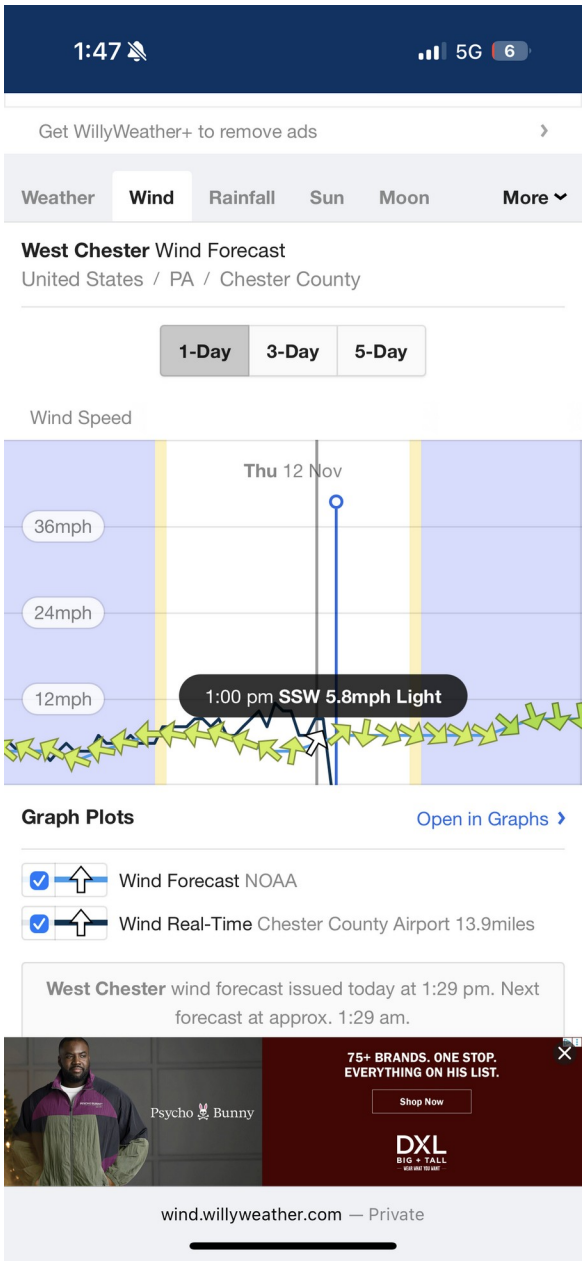


FIG. 20 1:47 PM — Wind forecast 2 PM. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WillyWeather / screenshot	IMG_6564.png	SHA-256 3815c9b9fd4efee31e3b…

CONTEXTUAL EXHIBIT / EX-014 / ORIGINAL ASSET

1:47 PM · Wind forecast detail

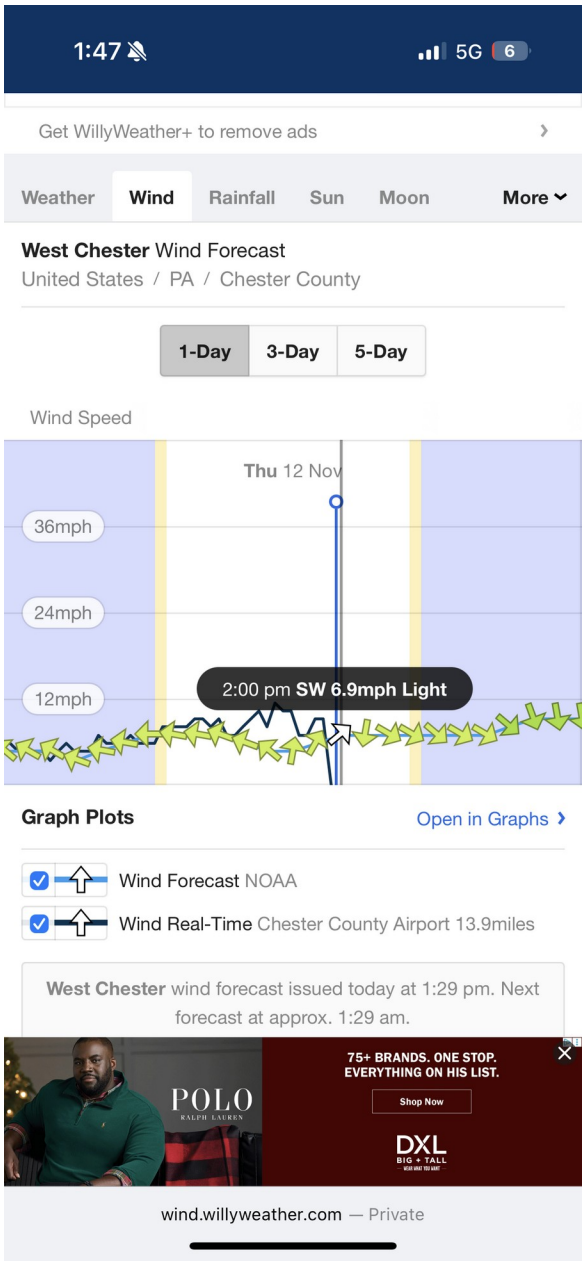


FIG. 21 1:47 PM — Wind forecast detail. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WillyWeather / screenshot	IMG_6565.png	SHA-256 17766ec92ad7b6f20cb2…

CONTEXTUAL EXHIBIT / EX-015 / ORIGINAL ASSET

1:35 PM · Regional radar gallery screenshot

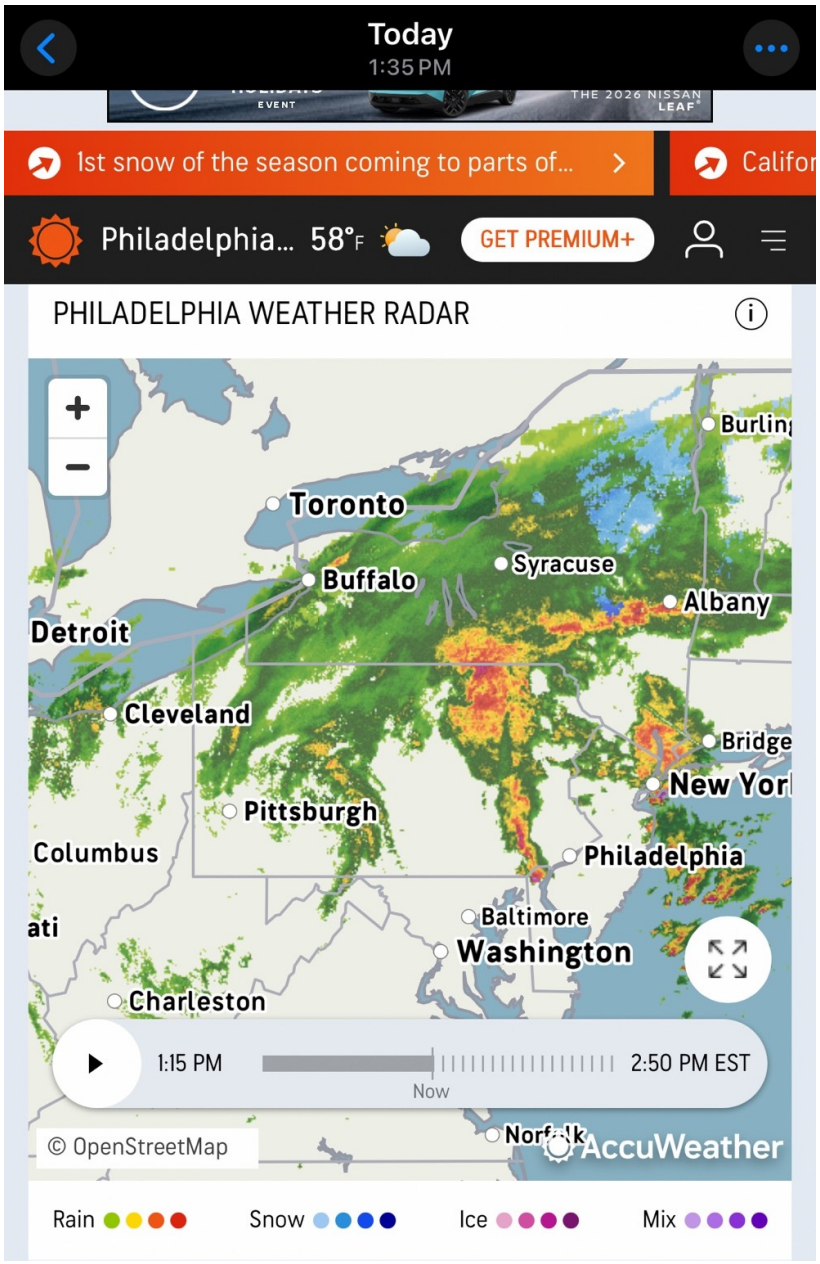


FIG. 22 1:35 PM — Regional radar gallery screenshot. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
AccuWeather / screenshot	IMG_6573.jpeg	SHA-256 a1d158c56388a2b1a266…

CONTEXTUAL EXHIBIT / EX-016 / ORIGINAL ASSET

1:36 PM · Regional radar gallery detail

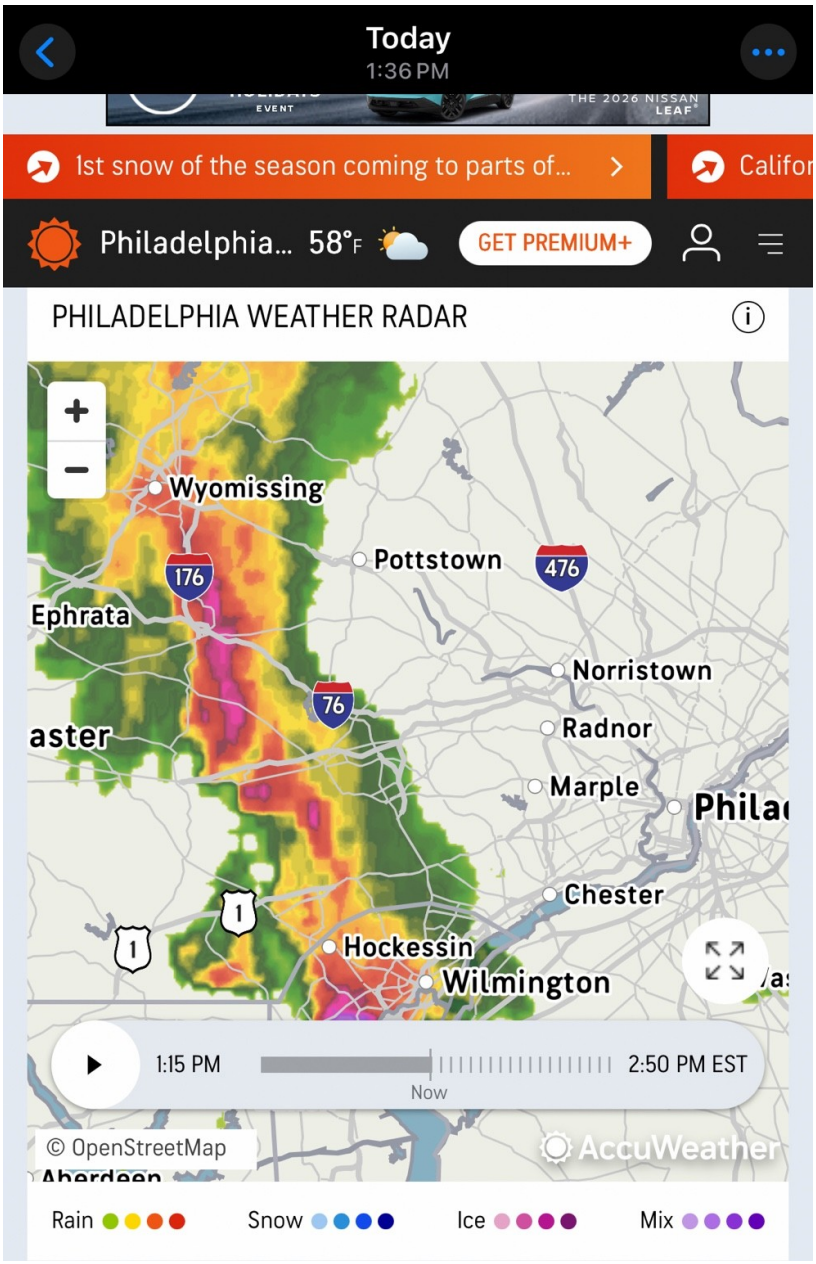


FIG. 23 1:36 PM — Regional radar gallery detail. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
AccuWeather / screenshot	IMG_6574.jpeg	SHA-256 90b75810df9771aa42fc…

CONTEXTUAL EXHIBIT / EX-017 / ORIGINAL ASSET

1:36 PM · Coatesville radar gallery detail

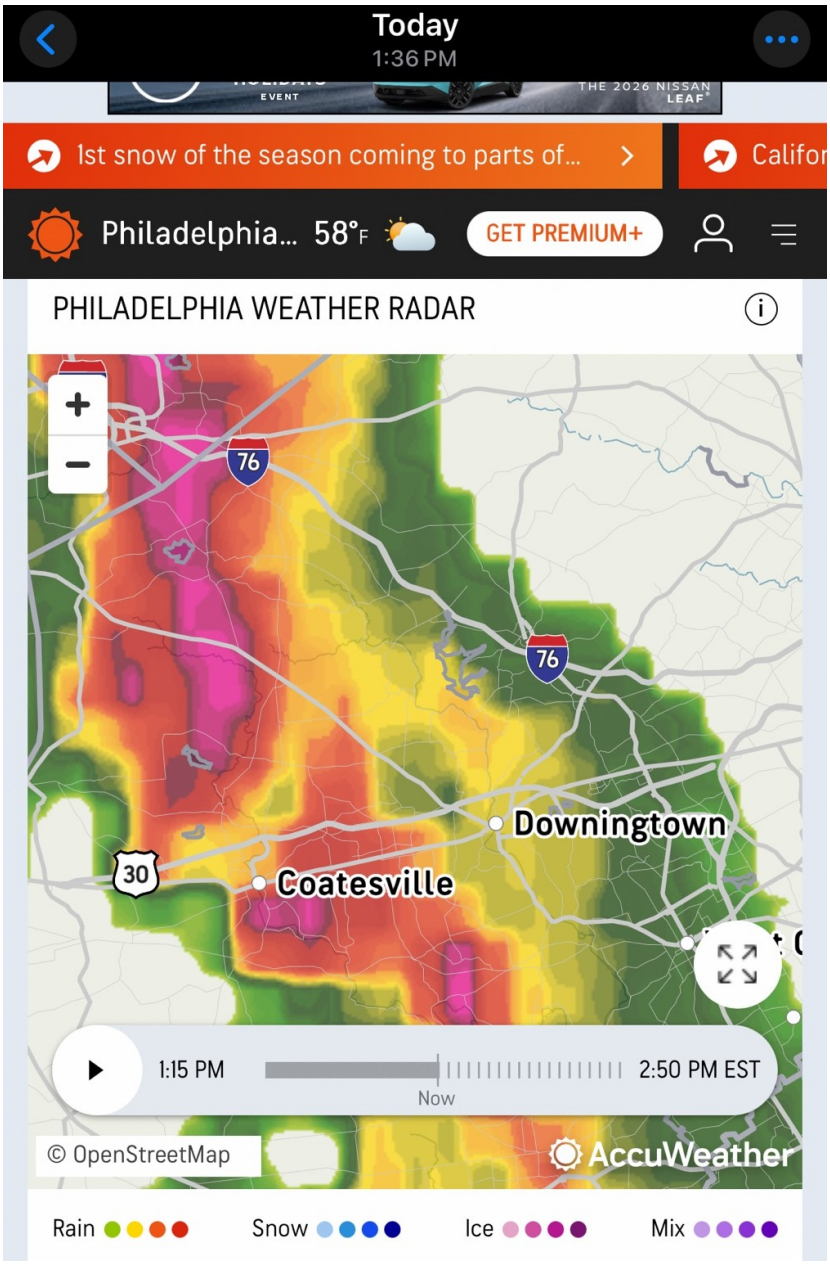


FIG. 24 1:36 PM — Coatesville radar gallery detail. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
AccuWeather / screenshot	IMG_6575.jpeg	SHA-256 53075d3532158287fbc4…

CONTEXTUAL EXHIBIT / EX-018 / ORIGINAL ASSET

1:38 PM · Tree canopy and sky



FIG. 25 1:38 PM — Tree canopy and sky. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User field image / gallery	IMG_6576.jpeg	SHA-256 be3648f88392b4c1d7c4…

CONTEXTUAL EXHIBIT / EX-019 / ORIGINAL ASSET

1:41 PM · Regional pressure gallery capture

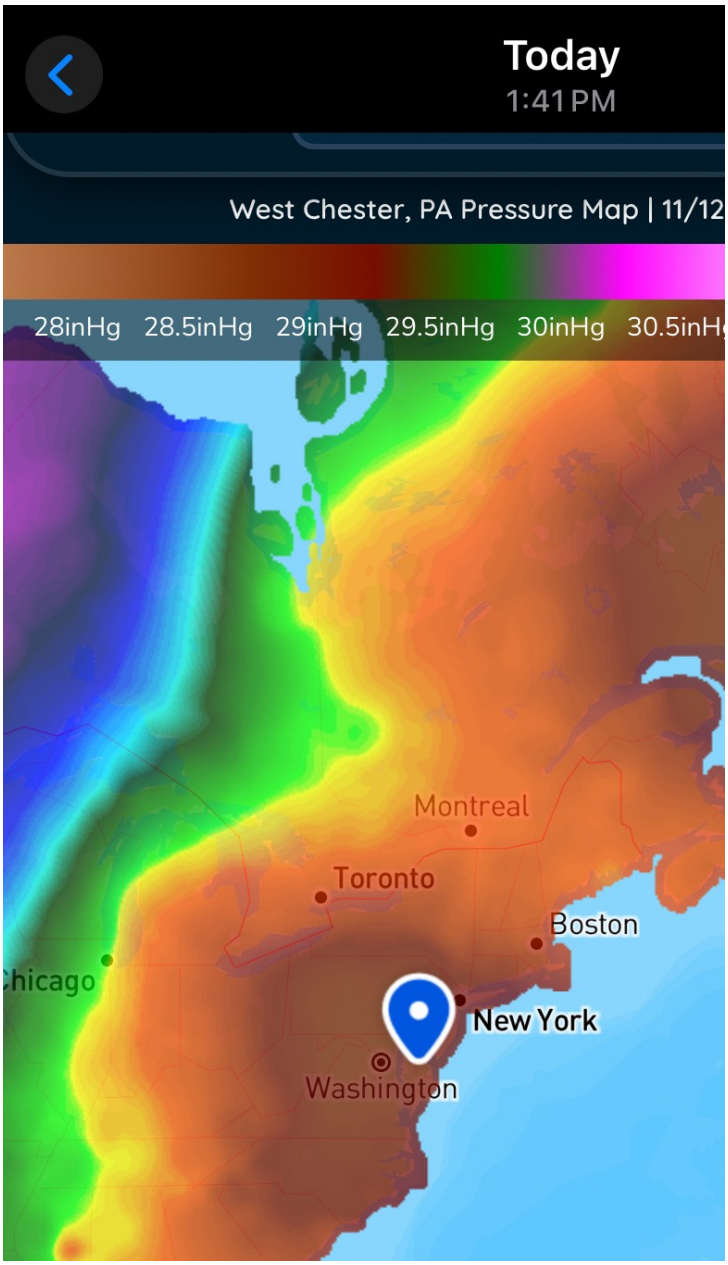


FIG. 26 1:41 PM — Regional pressure gallery capture. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6577.jpeg	SHA-256 78abf99d1aa240153749…

CONTEXTUAL EXHIBIT / EX-020 / ORIGINAL ASSET

1:41 PM · Philadelphia pressure gallery capture

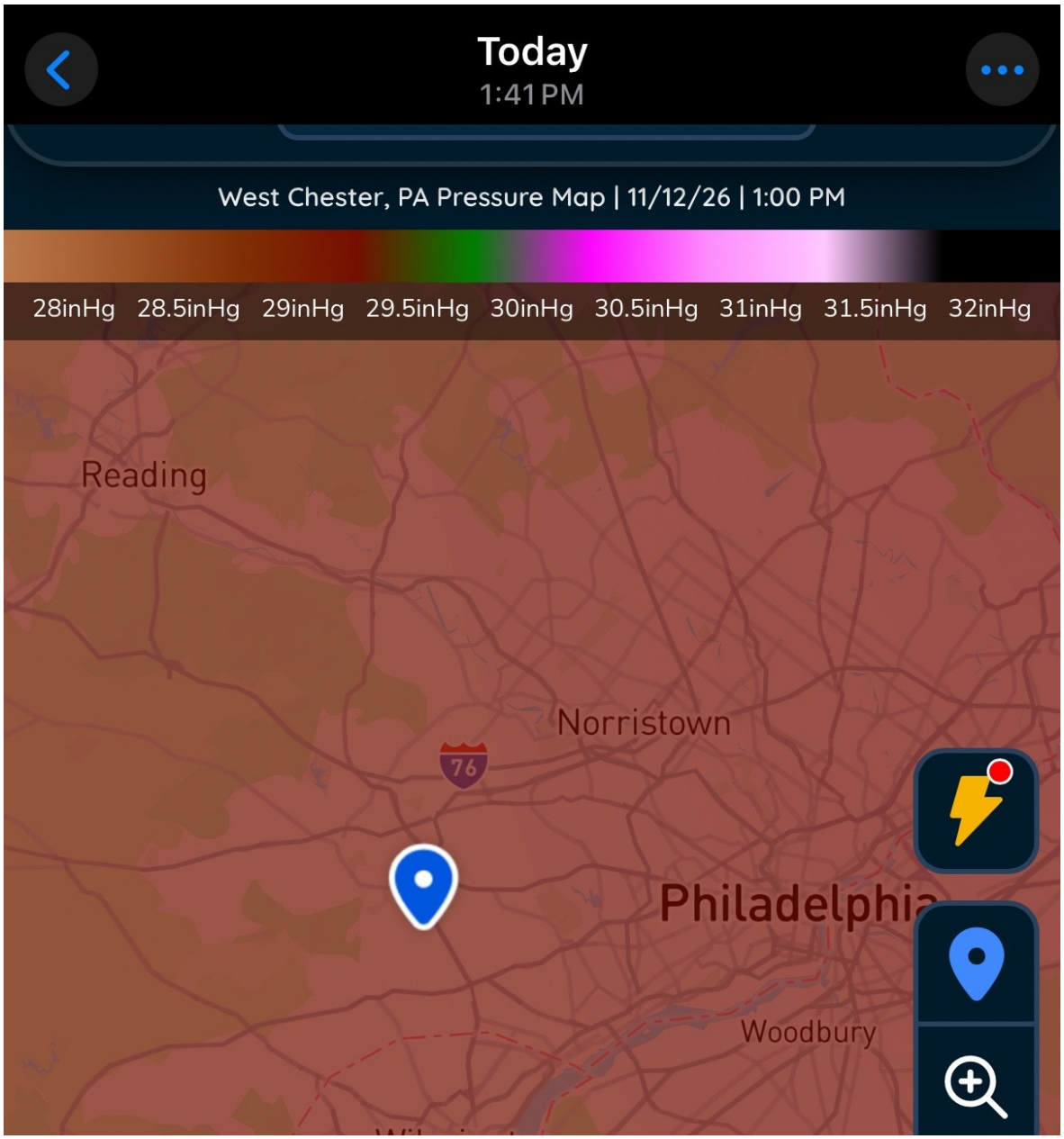


FIG. 27 1:41 PM — Philadelphia pressure gallery capture. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WeatherBug / screenshot	IMG_6578.jpeg	SHA-256 fd9d87e0e100e2f78b8f…

CONTEXTUAL EXHIBIT / EX-021 / ORIGINAL ASSET

1:43 PM · Tree and nest



FIG. 28 1:43 PM — Tree and nest. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User field image / gallery	IMG_6579.jpeg	SHA-256 6804df2abbe526f296c3…

CONTEXTUAL EXHIBIT / EX-022 / ORIGINAL ASSET

1:46 PM · Conifer foliage frame



FIG. 29 1:46 PM — Conifer foliage frame. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User field video / gallery	IMG_6580.jpeg	SHA-256 72ed92ff97cc61c21d21…

CONTEXTUAL EXHIBIT / EX-023 / ORIGINAL ASSET

1:47 PM · West Chester wind forecast detail

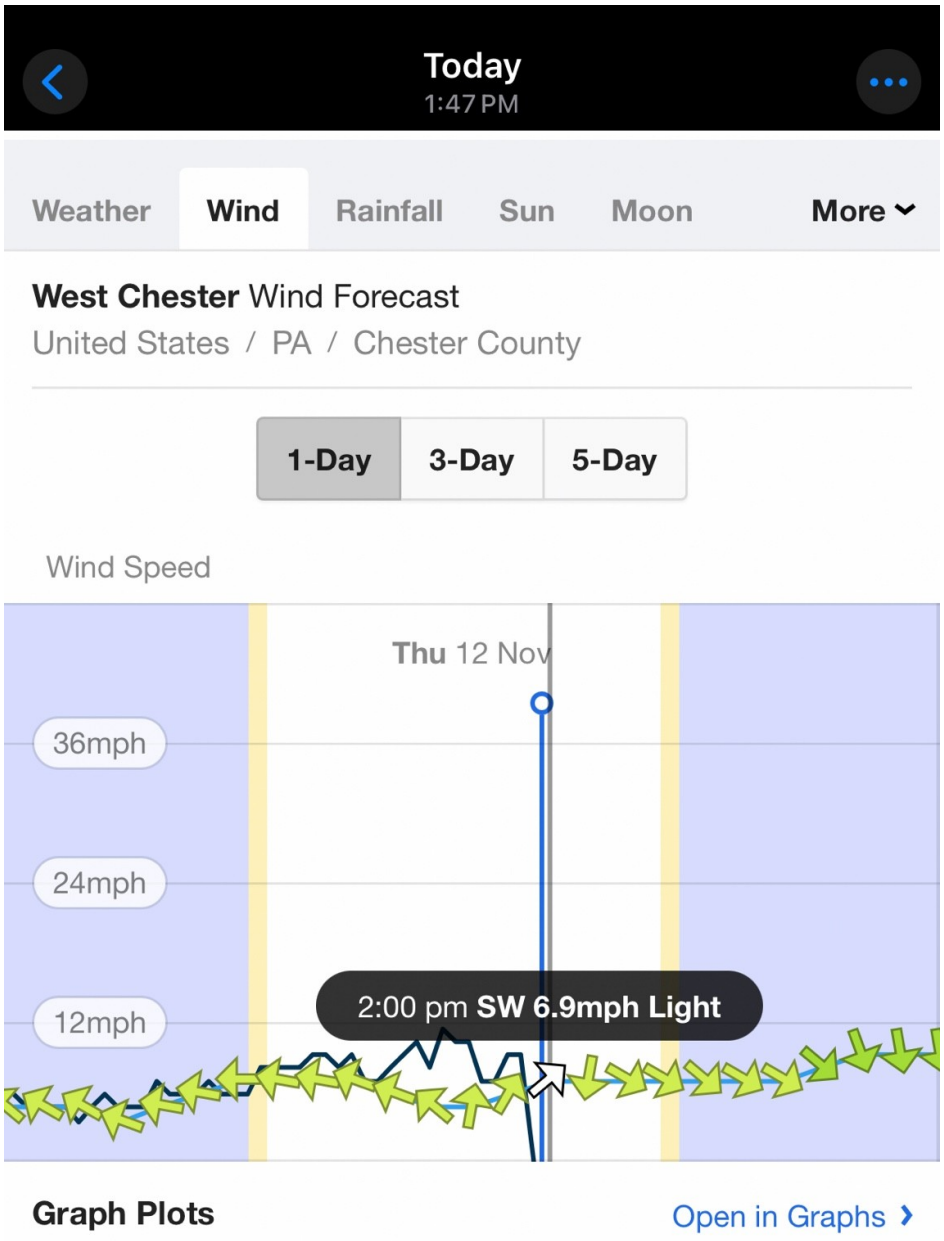


FIG. 30 1:47 PM — West Chester wind forecast detail. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
WillyWeather / screenshot	IMG_6581.jpeg	SHA-256 4f6754b3532038cadf47…

CONTEXTUAL EXHIBIT / EX-024 / ORIGINAL ASSET

storm period · Parking area rainfall and curb runoff



FIG. 31 storm period — Parking area rainfall and curb runoff. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User field photograph	IMG_BB2F708E-DDA7-4A67-98E2-4ED40775AE0D.jpeg	SHA-256 be7aa9f6e15f420e51e0…

CONTEXTUAL EXHIBIT / EX-025 / ORIGINAL ASSET

storm period · Tree base and sidewalk ponding



FIG. 32 storm period — Tree base and sidewalk ponding. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User field photograph	IMG_DD1D6951-06AF-4066-8D6F-516FFC7DD833.jpeg	SHA-256 faedca727e1231a6c2aa…

CONTEXTUAL EXHIBIT / EX-026 / ORIGINAL ASSET

storm period · Grassy slope drainage



FIG. 33 storm period — Grassy slope drainage. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
User field photograph	IMG_1C2F53A6-961F-47D0-AD6C-30D051A62524.jpeg	SHA-256 4cd42f6d04030de07293…

CONTEXTUAL EXHIBIT / EX-027 / ORIGINAL ASSET

Historical · Original StormWise tornado-potential screen

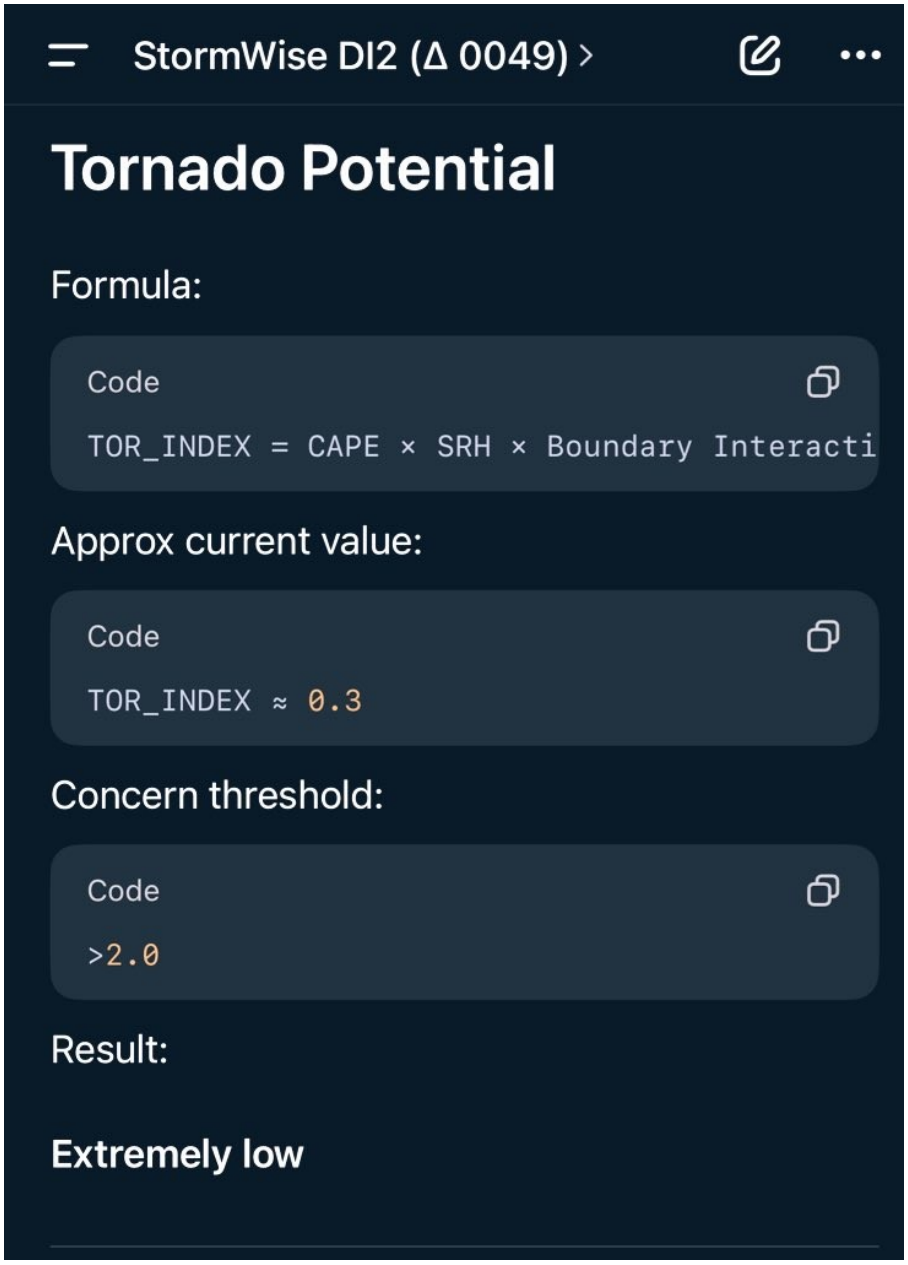


FIG. 34 Historical — Original StormWise tornado-potential screen. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
StormWise / user screenshot	IMG_9505.jpeg	SHA-256 33f3d6e406dafd48b543…

CONTEXTUAL EXHIBIT / EX-028 / ORIGINAL ASSET

Historical · Evidence-bounded entropy formula

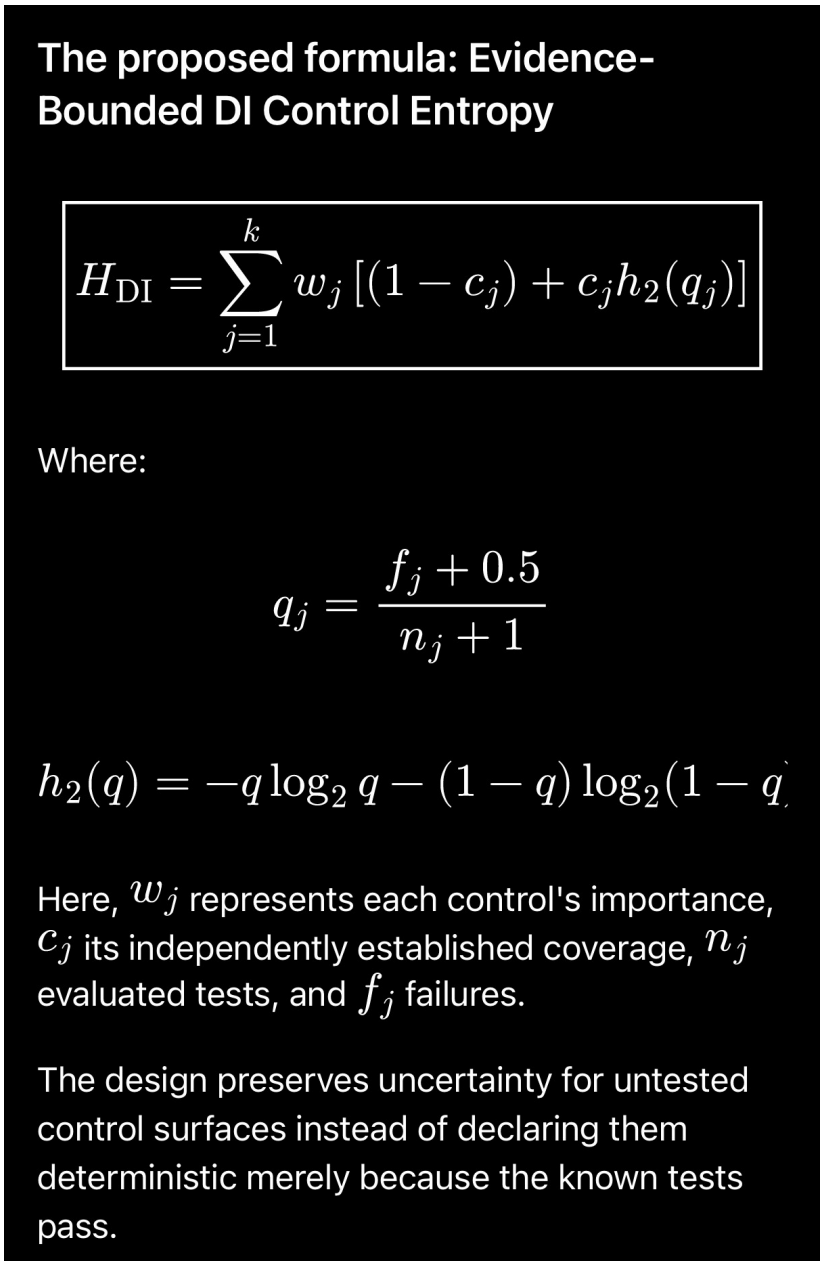


FIG. 35 Historical — Evidence-bounded entropy formula. Original uploaded image, unredacted.

Source	Original asset	Integrity reference
Research formula screenshot	IMG_3228.jpeg	SHA-256 0a831fe31ba47a7a04f6…

FIELD VIDEO ATLAS

33 / Foliage-motion clips

Three MP4 source clips are preserved unmodified in the evidence bundle. The following non-destructive contact sheets reproduce three frames per clip, each extracted at a stated offset. They are illustrative and do not constitute a calibrated anemometer or Doppler velocity product.

VIDEO EXHIBIT / V-01

1:47:44 PM · Foliage video A

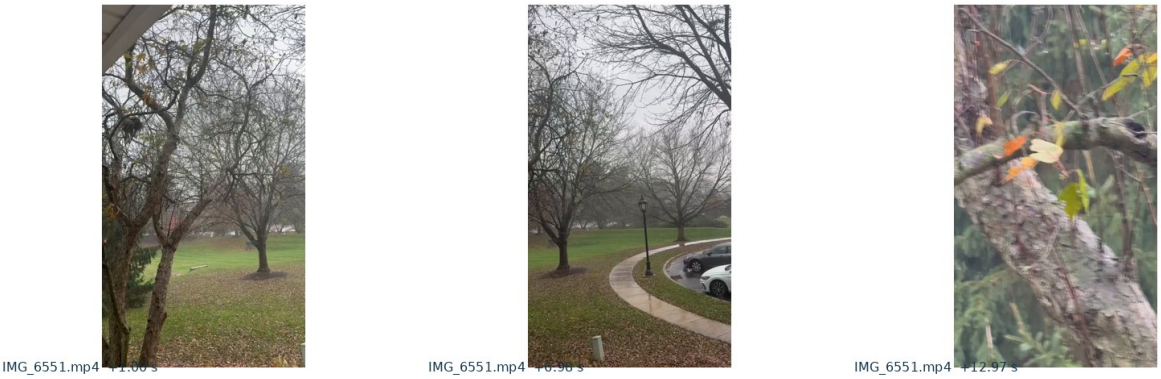


FIG. 36 Contact sheet sampled from IMG_6551.mp4; approximate source duration 13.97 seconds. Full MP4 retained without editing in evidence bundle.

Capture context	Technical note
Time label	1:47:44 PM EST (container creation timestamp shown in source receipt)
Frame offsets	Approximately 1.0 s, midpoint, and 1.0 s before end
Observational role	Visible foliage movement and camera motion in an outdoor scene
Analytical role	Contextual only; no absolute wind-speed estimate derived from pixel displacement
Source hash	7d79783350a47bfecb150c3d1962ad89238a7742a3515bc7b80fed1636cc9fcf

VIDEO EXHIBIT / V-02

1:47:44 PM · Foliage video B

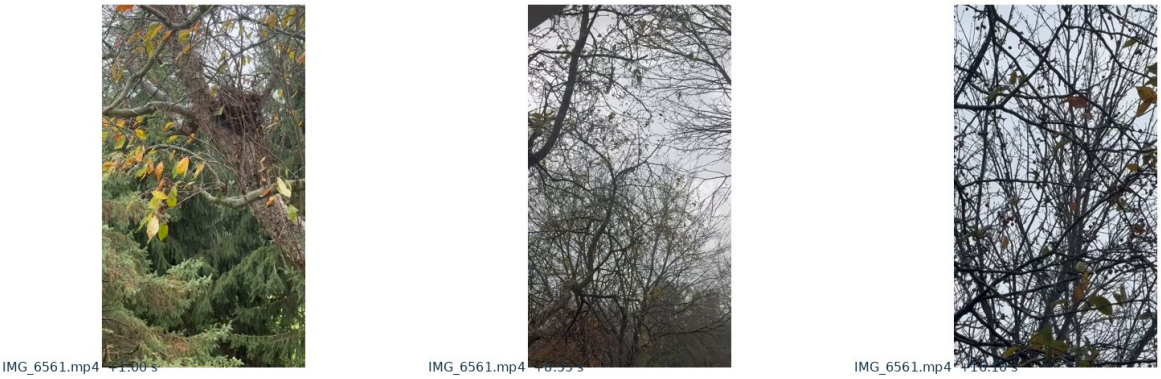


FIG. 37 Contact sheet sampled from IMG_6561.mp4; approximate source duration 17.10 seconds. Full MP4 retained without editing in evidence bundle.

Capture context	Technical note
Time label	1:47:44 PM EST (container creation timestamp shown in source receipt)
Frame offsets	Approximately 1.0 s, midpoint, and 1.0 s before end
Observational role	Visible foliage movement and camera motion in an outdoor scene
Analytical role	Contextual only; no absolute wind-speed estimate derived from pixel displacement
Source hash	0fda3ea72c574f6be101ed17a71bcd9696e6c41290623a0e53618d6f75df5da5

VIDEO EXHIBIT / V-03

1:48:13 PM · Foliage video C

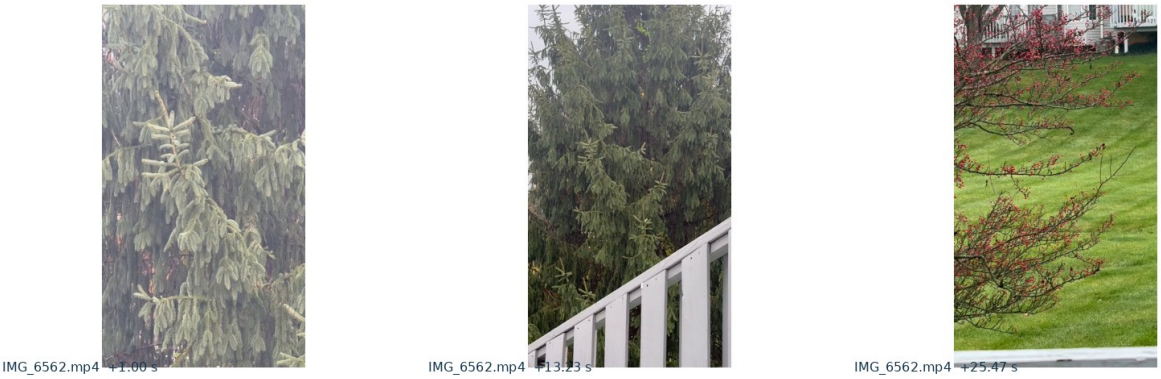


FIG. 38 Contact sheet sampled from IMG_6562.mp4; approximate source duration 26.47 seconds. Full MP4 retained without editing in evidence bundle.

Capture context	Technical note
Time label	1:48:13 PM EST (container creation timestamp shown in source receipt)
Frame offsets	Approximately 1.0 s, midpoint, and 1.0 s before end
Observational role	Visible foliage movement and camera motion in an outdoor scene
Analytical role	Contextual only; no absolute wind-speed estimate derived from pixel displacement
Source hash	08c32a36fa3a4bb548945e924dfbf388c49833f4c059c5e54b1bb333bb20a6c7

REPLAY / CODE / INDEPENDENT REVIEW

34 / Numerical replay protocol

The independent calculation script reads the structured book receipt and repeats the pressure conversion, wind changes, compass angular difference, gust factor and DI-TOR-01 synthetic test with high-precision decimal arithmetic.

Step	Independent analyst action
1	Obtain the original KOQN raw METAR strings included in the receipt and, when possible, compare them with the NOAA/AWC-derived archive.
2	Identify pressure groups A2954, A2944, A2938, wind groups and timestamps in the raw lines. Preserve UTC/EST time mapping.
3	Run the accompanying Python 3 standard-library replay program against the book receipt JSON.
4	Verify pressure outputs −1.8060739556 and −2.0318332000 hPa/h; airport wind +2 kt; forecast wind +1.1 mph.
5	Verify the synthetic E01 = 0.3125 and T_A = 0.15625; do not label either an actual event score.
6	Compare any claimed new event score against verified MLCAPE, signed SRH, storm motion, calibrated B and a versioned formula.
7	Treat changes in event inputs, source records or model assumptions as a new version; preserve old receipts.

Execution command

python DI_Weather_Station_Inside_the_Watch_Replay.py

Script dependencies: Python 3.9 or newer, standard library only. The script prints a JSON object with full decimal outputs and contains regression assertions. For external audit the parent report, original receipt, full asset files and their cryptographic hashes are preserved independently.

Formal release position

Gate	Status
Event-data arithmetic	PASS for verified KOQN and forecast comparisons
Formula arithmetic	PASS for synthetic test vector
DI-TOR-01 event measurement	UNRESOLVED
DI-TOR-01 predictive validation	NOT EXECUTED
GOEI / evidence-bounded control entropy	UNRESOLVED

Gate	Status
WeatherReady	NOT VALIDATED
AlertRelease	BLOCKED
Model warning authority	NONE

SOURCE BIBLIOGRAPHY / ASSET MANIFEST

35 / Documents, authorship and original files

The source library combines official meteorological products, a preserved secondary airport archive, research methodology and contemporaneous visual evidence. This selectively redacted public edition retains the evidence and original-source classifications. The unchanged internal research master, executable original receipt and confidential implementation are not part of this release.

Technical and official source links

S01 KOQN METAR sequence, Nov 12 2026. https://runwaypanel.com/airports/KOQN/metar-history?year=2026

S02 NWS Mount Holly AFD, 02:41 AM EST Nov 12. https://forecast.weather.gov/product.php?format=CI&highlight=o&issuedby=PHI&product=AFD&site=NWS&version=1

S04 NWS Chester County watch status. https://www.weather.gov/phi/

S05 SPC November 12 Slight Risk update. https://www.reddit.com/r/NYCmeteorology/comments/1xuii05/nws_spc_1120_am_thursday_november_12_2026_day_1/

S06 Tornado Watch 705 text. https://www.reddit.com/r/TornadoWatch/comments/1xujw3r/nov_12_2026_tornado_watch_number_705_issued_at/

S11 NWS/NOAA near-storm environment parameter review. https://www.weather.gov/media/hun/research/NSEParameters.pdf

Preserved digital sources

The companion manifest lists the original filenames, byte sizes, SHA-256 hashes and asset classes for 39 event-related files, including the earlier four-page StormWise scan, its original JSON receipt, the DI Weather Station formula/runtime references, all indexed photographs and screenshots, and the three original MP4 videos.

The machine-readable calculation receipt binds the copied raw KOQN observations, every cited numeric value, the formula definitions, case-status controls and the photo-exhibit registry into one reproducible record. Its identifiers permit a later public edition to be derived without changing the internal original.

FINAL RESEARCH POSITION

The verified analysis is a source-bound study of a real watch-period severe-weather event. At 1:30 PM the inputs support the official watch and independent atmospheric-pressure/wind arithmetic. They do not support an actual numerical DI-TOR-01 result, a self-issued tornado warning, an all-clear, or a claim of scientific predictive validation.